

BEST 'A' LEVEL BIOLOGY REVISION NOTES[®]

TOPIC: **TRANSPORT IN HUMANS**

Compiled by **G. TARUVINGA**

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TRANSPORT SYSTEMS IN HUMANS

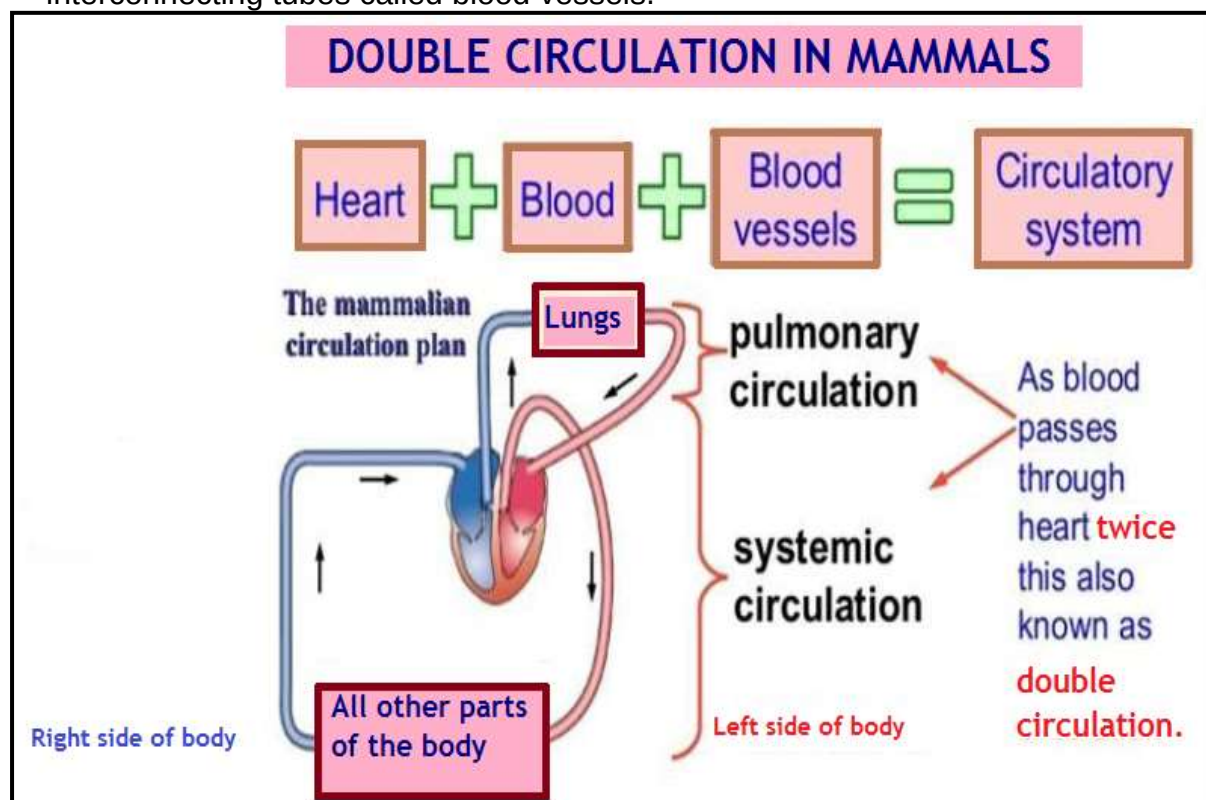
TOPIC 8.11.1 MAMMALIAN CIRCULATORY SYSTEM

TRANSPORT IN MAMMALS

CONCEPT	OBJECTIVES Learners should be able to:	CONTENT (ATTITUDES, SKILLS AND KNOWLEDGE)	SUGGESTED LEARNING ACTIVITIES AND NOTES	SUGGESTED RESOURCES
8.11.1 Mammalian circulatory system	1) identify arteries, veins and capillaries 2) explain the role of haemoglobin in the transportation of oxygen and carbon dioxide 3) explain the Bohr effect 4) explain the significance of the difference in the affinity for oxygen between: – Haemoglobin and myoglobin – Maternal and foetal haemoglobin 5) describe the cardiac cycle 6) explain how heart action is initiated and controlled 7) explain the meaning of the terms systolic blood pressure, diastolic blood pressure and hypertension 8) discuss the long term consequences of exercise on the cardiovascular system	- Arteries, veins and capillaries - Transportation of oxygen and carbon dioxide - Oxygen dissociation curves - Difference in oxygen affinity between: i. Haemoglobin and myoglobin ii. Maternal and foetal haemoglobin - Cardiac cycle - Pacemaker - Myogenic control - Systolic and diastolic blood pressure - Hypertension - Improved cardiac output - Normal resting pulse rate - Efficient cardiovascular system	• Recognising the vessels under the light microscope. • Drawing plan diagrams of blood vessels. • Discussing the transportation of oxygen and carbon dioxide. • Analysing oxygen dissociation curves. • Discussing the differences in oxygen affinity. • Observing cardiac cycle simulations. • Observing heart initiation simulations. • Measuring blood pressure. • Analysing the results. • Discussing the long term consequences of exercise.	• Microscope • Prepared slides • Photomicrographs • ICT tools • Print media • Braille software/Jaws • Heart models • Sphygmomanometer • Stethoscope • Research tools

The mammalian circulatory system

- It is made up of a pump called the heart, a fluid called blood, a system of interconnecting tubes called blood vessels.



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Mammalian circulation is Double circulation

- A **double** circulatory system refers to a system in which blood pumps through the heart **twice** during each trip around the body.
- This means that we have **two loops/circuits** in our body in which blood circulates. One carries oxygenated blood meaning oxygen rich blood and the other carries deoxygenated blood which means it has little to no oxygen but a lot of carbon dioxide.
- The **two loops** are the **systemic circulation** and the **pulmonary circulation**.

Systemic circulation

- **Systemic blood circulation carries oxygenated blood away from the heart to the body and returns deoxygenated blood back to the heart.**
 - Blood is pumped out of the left ventricle into the aorta and travels from there to all parts of the body except the lungs.
 - It returns to the right side of the heart in the vena cava.
 - This is called the systemic circulation.

Pulmonary circulation

- **Pulmonary circulation carries oxygen depleted blood away from the heart to the lungs and returns oxygenated blood back to the heart.**
 - It transports oxygen-poor blood from the right ventricle to the lungs where blood picks up a new blood supply.
 - Then it returns the oxygen-rich blood to the left atrium.

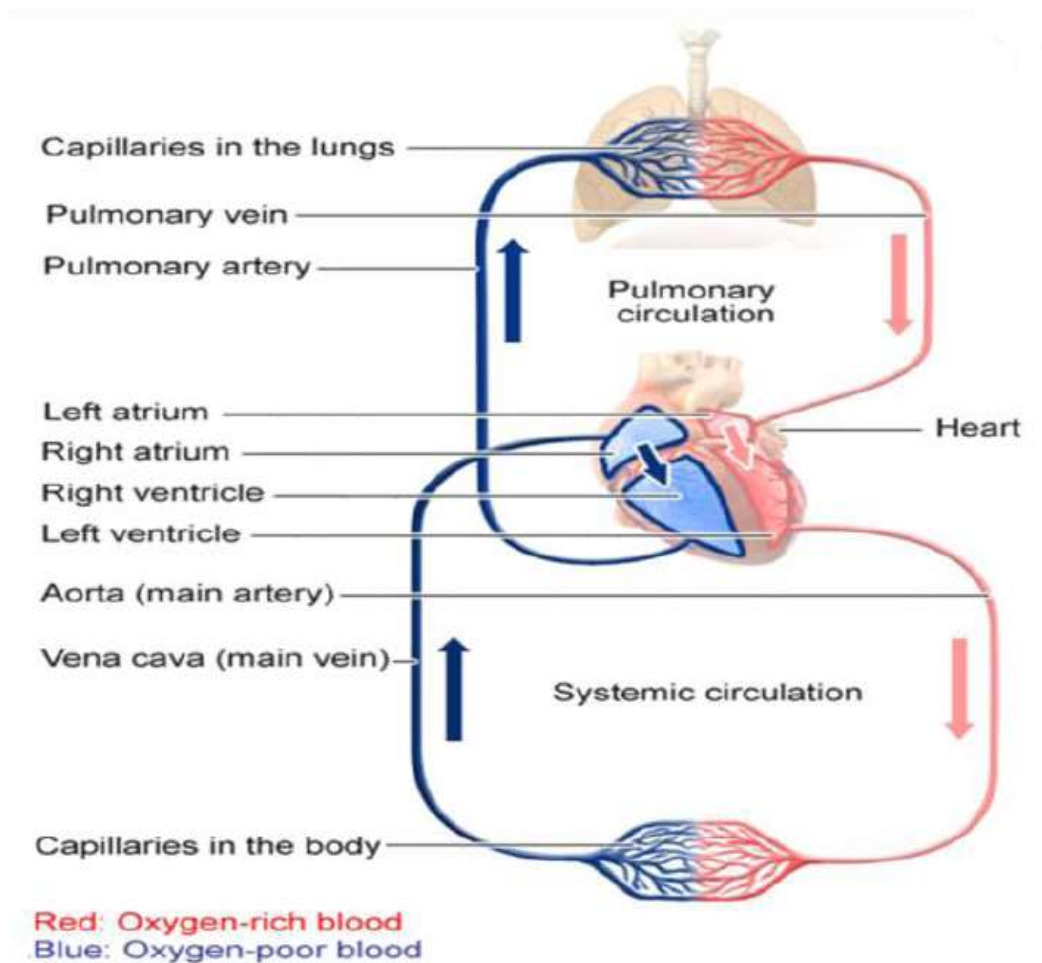
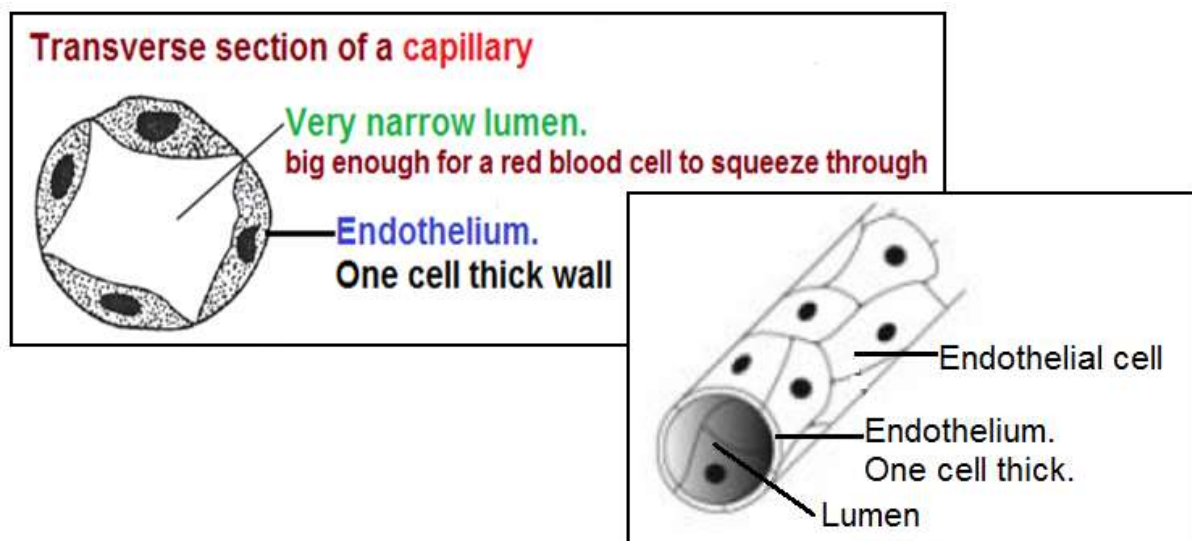
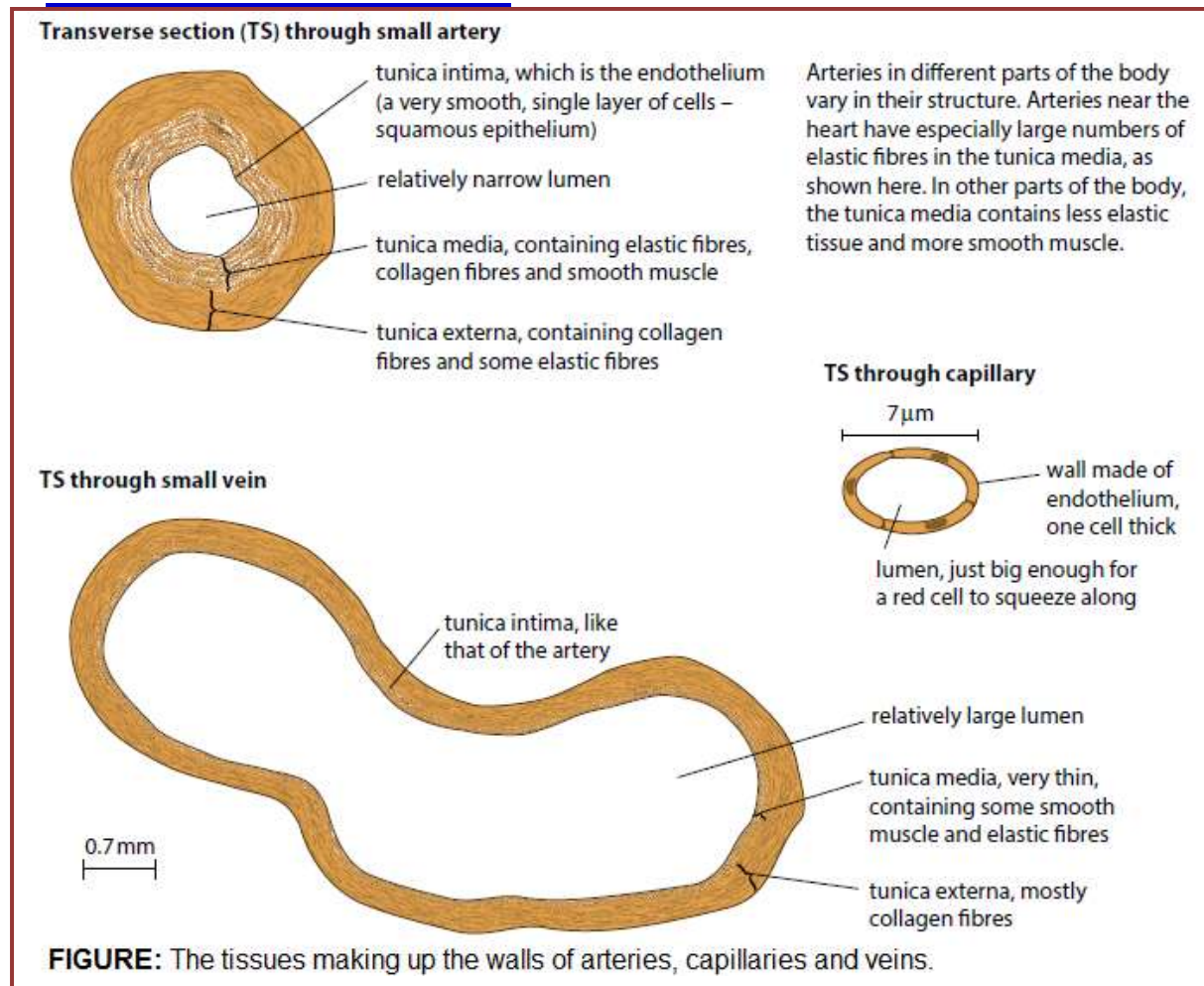


DIAGRAM: Systemic and pulmonary circulations.

BLOOD VESSELS

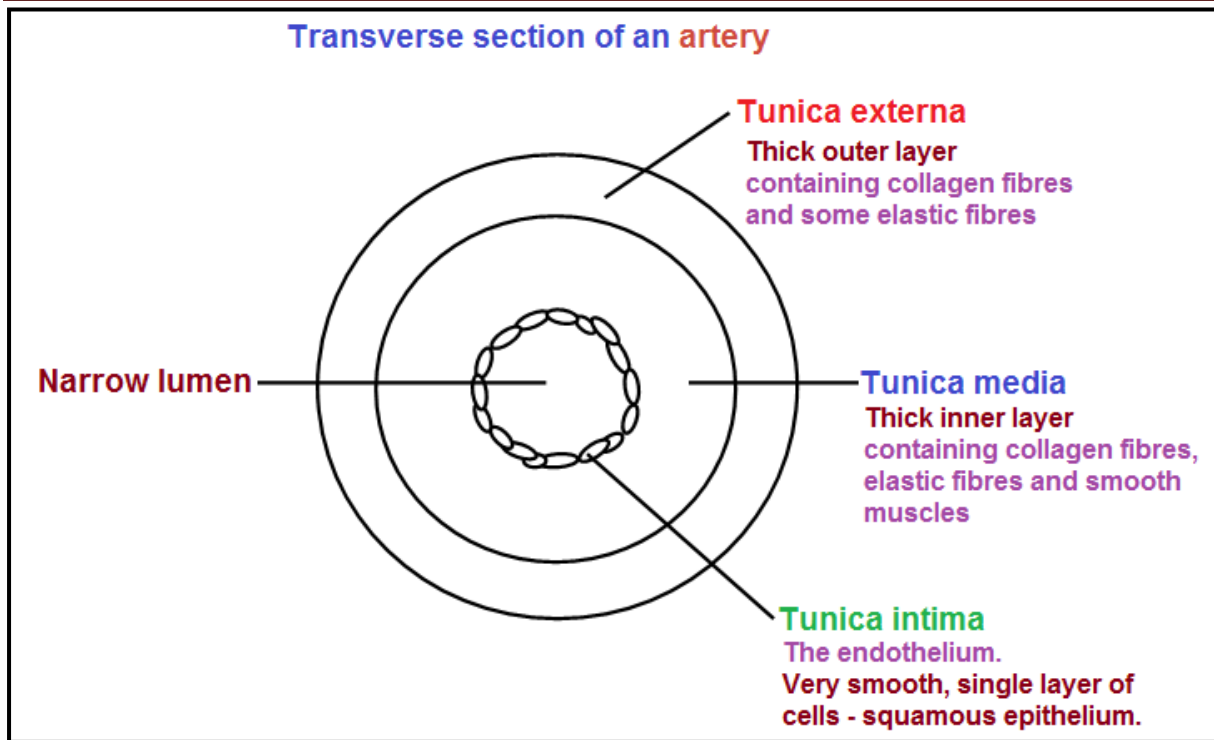
- The vessels making up the blood system are of **three** main types.
- **Vessels carrying blood away from the heart are known as arteries.**
- **Vessels carrying blood towards the heart are veins.**
- **Linking arteries and veins taking blood close to almost every cell in the body are tiny vessels called capillaries.**

Structure of blood vessels

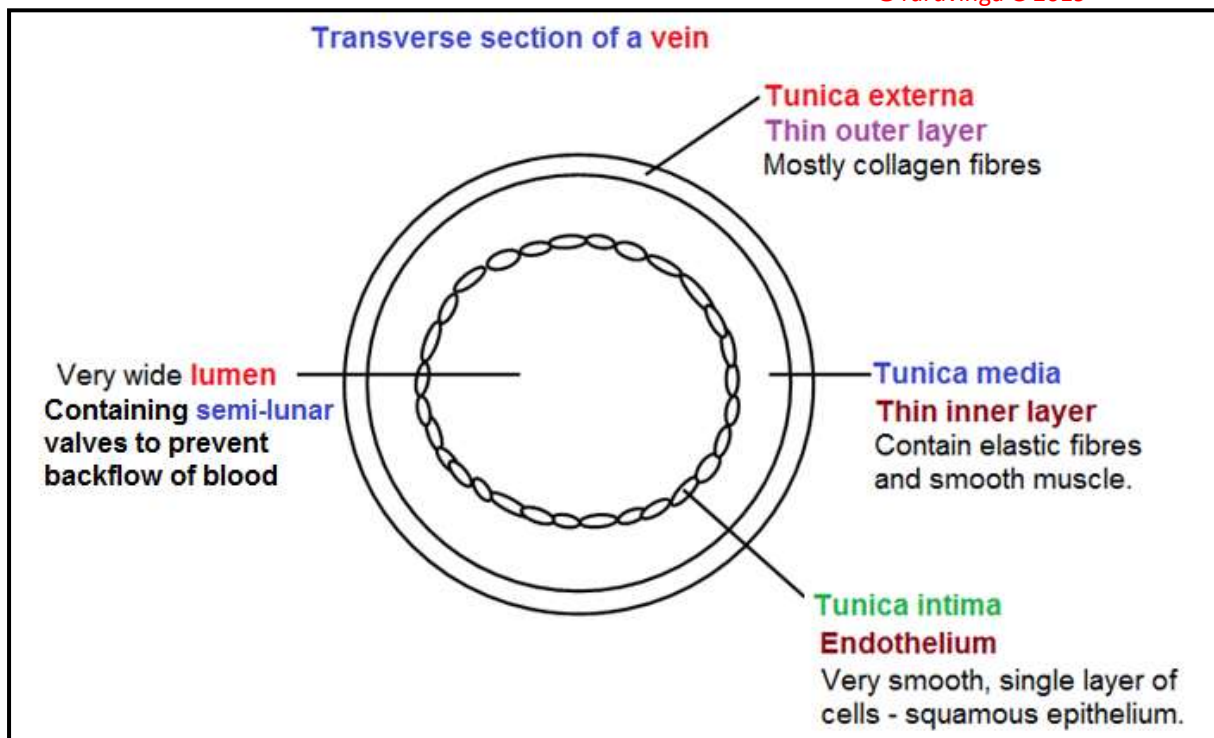


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TRANSPORT SYSTEMS IN HUMANS



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- Arteries and veins both have walls made up of **THREE** layers:
 - An inner endothelium (lining tissue) called the **tunica intima** made up of a layer of flat cells (squamous epithelium) fitting together like jigsaw pieces - this layer is **very smooth minimizing friction with the moving blood**.
 - A middle layer called the **tunica media** ('middle coat') containing smooth muscle collagen and elastic fibers.
 - An outer layer called the **tunica externa** ('outer coat') containing elastic fibers and collagen fibers.

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ARTERIES

- The function of **arteries** is to **transport blood away from the heart at high pressure to the tissues**.
 - All **systemic arteries** transport **oxygenated blood** at high pressure from left ventricle to body tissues except the **pulmonary artery** which transport **deoxygenated blood** at low pressure from right ventricle into lungs.
 - ❖ Blood that flows through arteries from the heart is **pulsing** and at a **high pressure**.
 - ❖ Therefore, arteries have **strong, thick, elastic** walls which can expand and recoil as the **high pressure blood** pulses through.
- The wall of the artery contains large amounts of **elastic fibers**. These allow the **wall to stretch** as pulses of blood surge through at **high pressure**.
- The **strength** and **elasticity** of artery walls is important in allowing them to **stretch and recoil** so that they will **not burst**.
- The artery walls stretch as the high-pressure blood surges into them and then **recoil inwards as the pressure drops**.
- Therefore as blood at high pressure enters an artery the artery becomes wider reducing the pressure a little as blood at lower pressure enters an artery - the artery wall recoils inwards giving the blood a small 'push' and raising the pressure a little.

Arterioles

- An arteriole is a small diameter blood vessel that extends and branches out from an artery and leads to capillaries.
- The walls of arterioles are similar to those of arteries but they have a greater proportion of smooth muscle.
- This muscle can contract narrowing the diameter of the arteriole and so reducing blood flow.
- This helps to control the volume of blood flowing into a tissue at different times.
- For example during exercise arterioles that supply blood to muscles in your legs would be dilated as their walls relax while those carrying blood to the gut wall would be constricted.

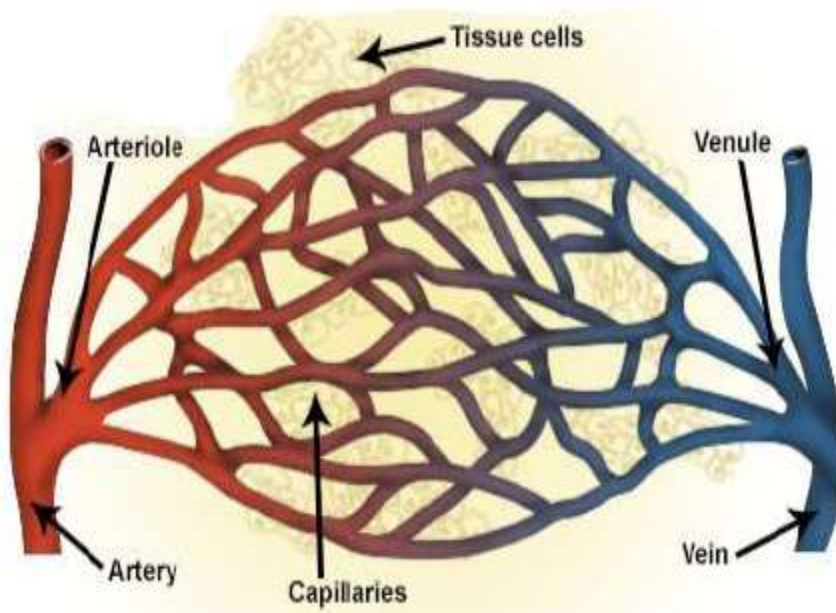


DIAGRAM: An artery branches into arterioles, which branch repeatedly into capillaries, which rejoin into venules which finally join into a vein.

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Artery → arterioles → capillaries → venules → vein.

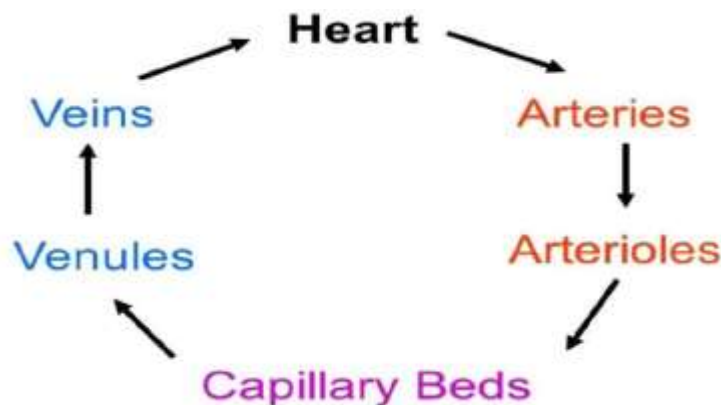
- Blood pressure drops significantly as blood flows through the arteries.
- As blood enters an arteriole it may have a pressure of 85 mmHg (11.3 kPa) but as it leaves and flows into a capillary then pressure will have dropped to about 35 mmHg (4.7 kPa).



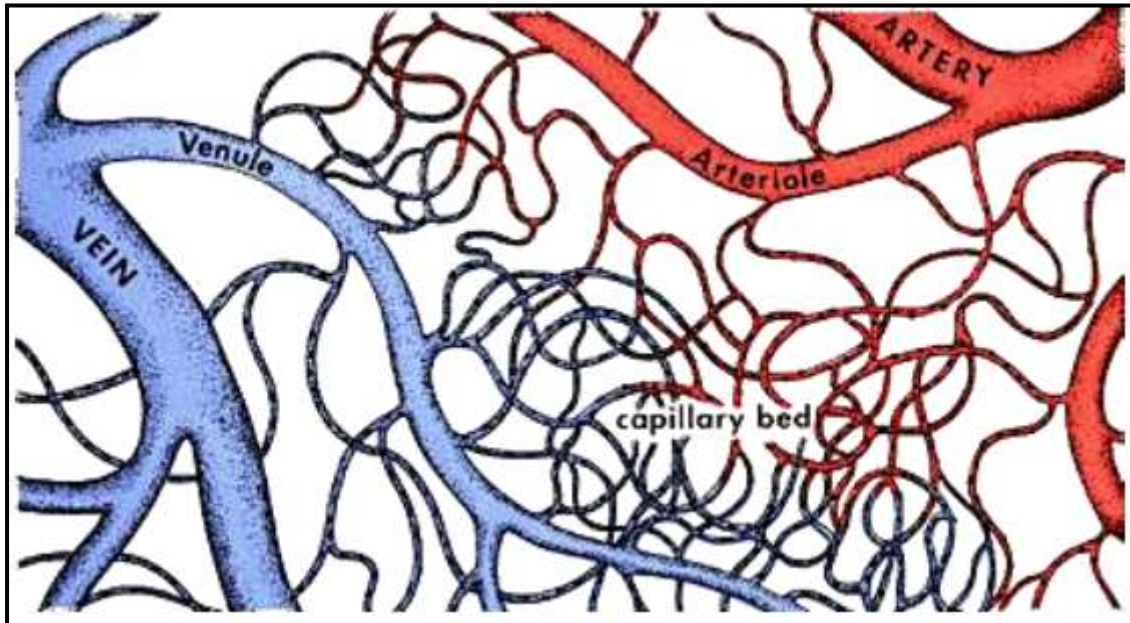
Photomicrograph of an artery (left) and a vein (right) ($\times 110$).

Capillaries

- The arterioles themselves continue to branch eventually forming the tiniest of all blood vessels capillaries.
- Capillaries form a network throughout every tissue in the body except the cornea and cartilage. Such networks are sometimes called **capillary beds**.



- **The function of capillaries is to take blood as close as possible to all cells allowing rapid exchange of substances between cells and blood.**
 - Capillaries are tiny vessels with just enough space for red blood cells to squeeze through.
 - Their walls are only **one cell thick**, and **have gaps** through which plasma (the liquid component of blood) can **leak out** to deliver nutrients, hormones and other requirements to body cells, and take away their waste products.
 - **Small size and thin walls reduce diffusion distance**, enabling exchange to take place rapidly between the blood and the body cells.



Veins

- **Venules** are small blood vessels that **connect the capillary beds to the veins**.
 - The capillaries gradually join with one another forming larger vessels called **venules**. These join to form veins.
 - **Artery** → **arterioles** → **capillaries** → **venules** → **vein**.

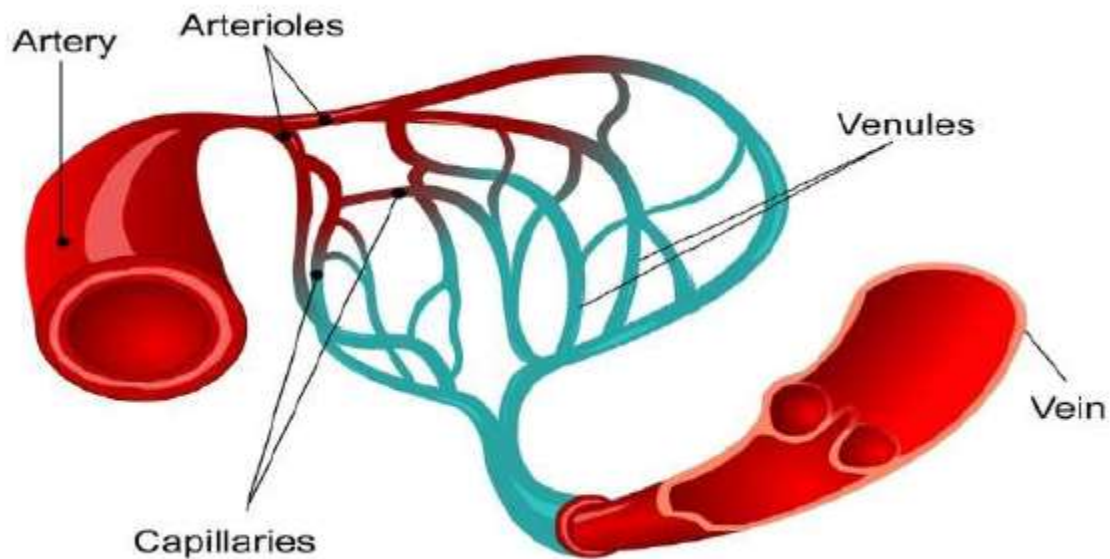
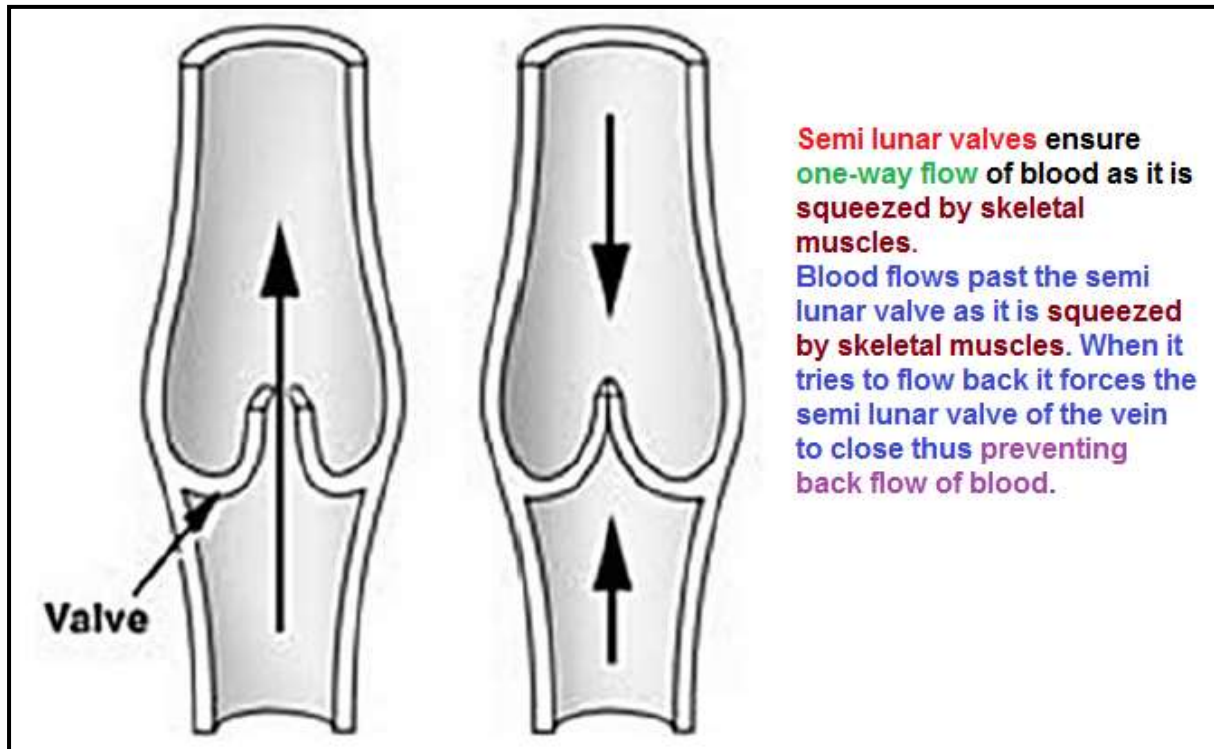


DIAGRAM: An artery branches into arterioles, which branch repeatedly into capillaries, which rejoin into venules which finally join into a vein.

- The **function** of **veins** is to **return low-pressure blood to the heart**.
 - All **systemic veins** **transport low-pressure deoxygenated blood towards the heart** except **pulmonary veins** which transport oxygenated blood from **lungs to left atrium**.
 - The walls of veins do not need to be as tough or as elastic as those of arteries as the blood is not at high pressure and is not pulsing.
 - The **lumen is larger** than in arteries, **reducing friction** which would otherwise slow down blood movement.

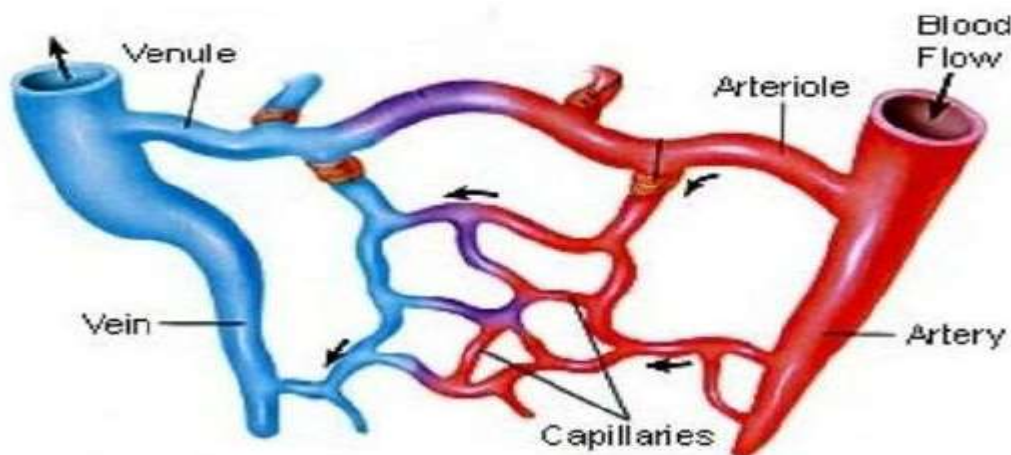
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- The lumen contain valves, to ensure that the blood **does not flow back** the wrong way.
- **Blood is kept moving** through many veins, for example those in the legs, **by the squeezing effect due to the contraction of the body/skeletal muscles** close to them.



How Blood Travels thru Vessels

heart → artery → arteriole → capillary → venule → vein → heart

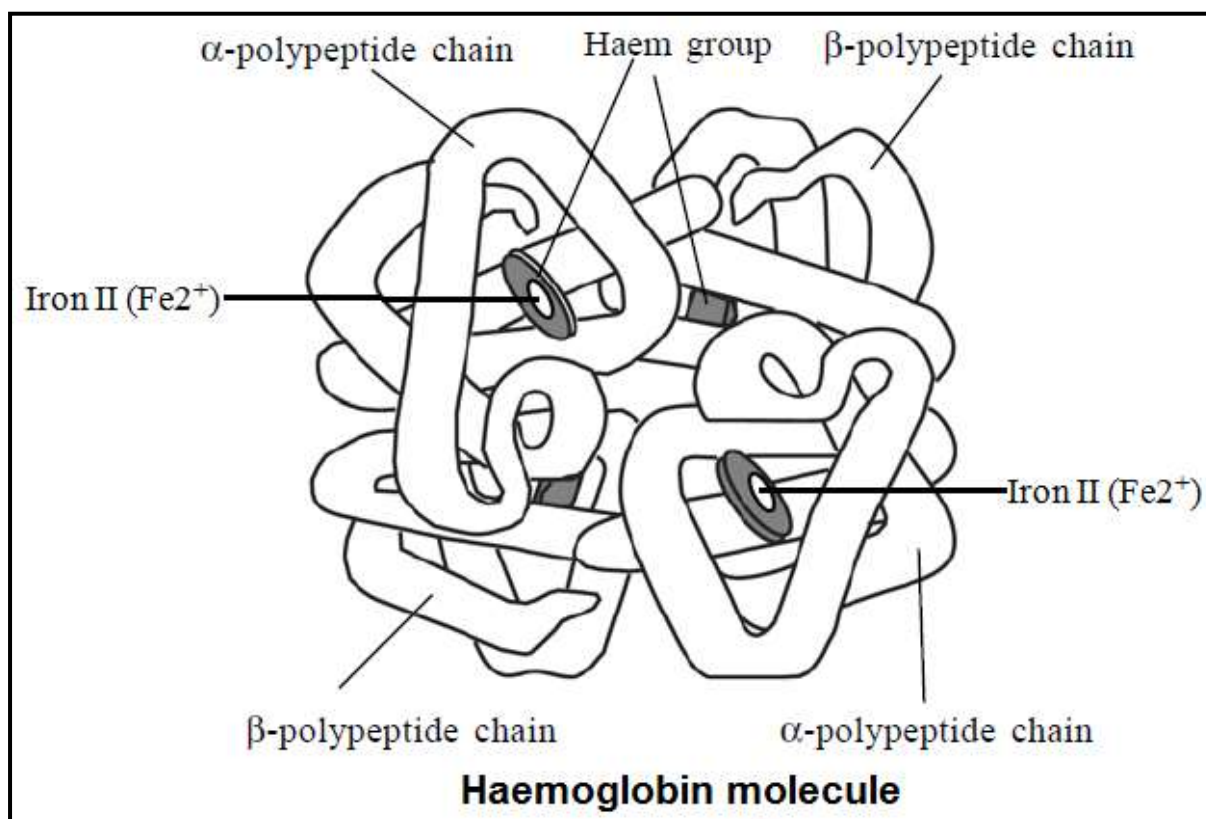


Haemoglobin

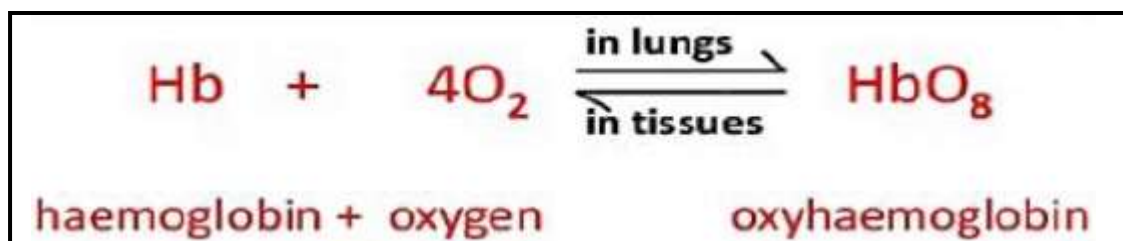
- A major role of the cardiovascular system is to transport oxygen **from the gas exchange surfaces of the alveoli in the lungs to tissues** all over the body.
- Body cells need a **constant supply** of oxygen in order to be able to carry out aerobic respiration.

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- Oxygen is transported around the body inside red blood cells in combination with the protein haemoglobin.
- Haemoglobin is the oxygen – carrying pigment in red blood cells.
- Each **hemoglobin molecule** is a globular protein made up of **four polypeptides** (two identical **α chains** and identical two **β chains**), each containing one **haem group**.



- Each haem group can combine with one oxygen molecule O_2 .
- Then each hemoglobin molecule can combine with four oxygen molecules (eight oxygen atoms):



The oxygen dissociation curve (ODC) for haemoglobin

- A molecule whose function is to transport oxygen from one part of the body to another must be able not only to pick up oxygen at the lungs, but also to release oxygen within **respiring** tissues.
- Hemoglobin performs this task very well.
- The **oxygen dissociation curve for haemoglobin (ODC)** is an important **tool for understanding how our blood carries and releases oxygen**.
- What is the oxygen dissociation curve? A graph that describes the relationship between the **partial pressure of oxygen** (x-axis) and **oxygen saturation** (y-axis).
- The **oxygen dissociation curve (ODC)** is sometimes called the **oxyhaemoglobin dissociation curve (OHDC)**.

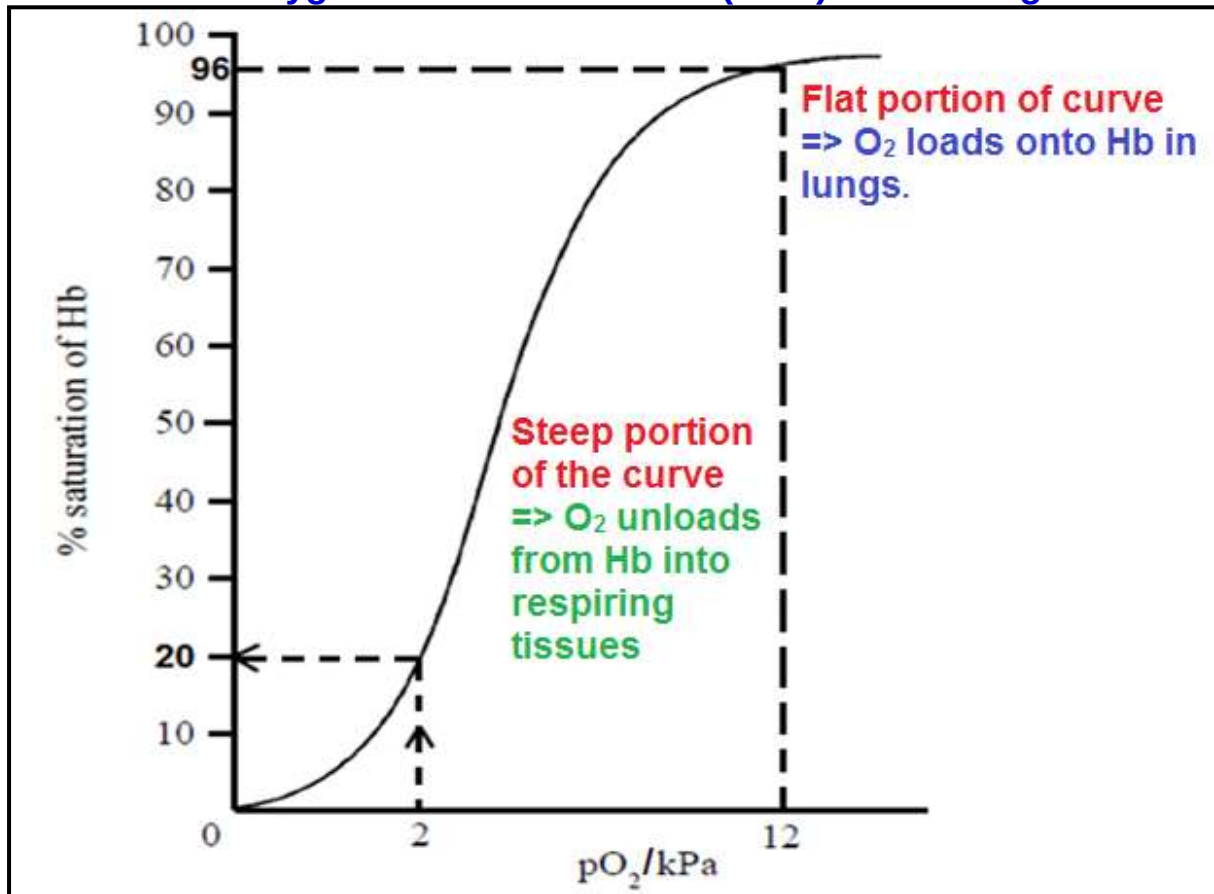
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- Specifically, the hemoglobin dissociation curve relates **oxygen saturation** and **partial pressure of oxygen (pO_2)** in the blood.
- Also it determines by what is called **haemoglobin's affinity for oxygen** how readily haemoglobin acquires and releases oxygen molecules from its surrounding tissue.
- The curve shows that at **low partial pressures of oxygen**, **the percentage saturation of haemoglobin is very low** – that is, **the haemoglobin is only combined with very little oxygen**.
- At **high partial pressures of oxygen** **the percentage saturation of hemoglobin is very high**; that is, **the haemoglobin is combined with large amounts of oxygen**.

TABLE: The varying ability of haemoglobin to carry oxygen

Partial pressure of oxygen / kPa	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Percentage saturation of haemoglobin	8.5	24.0	43.0	57.5	71.5	80.0	85.5	88.0	92.0	94.0	95.5	96.5	97.5	98.0

GRAPH: The oxygen dissociation curve (ODC) for haemoglobin



GRAPH: ODC for Hb. Steep portion of the curve => O_2 unloads from Hb into respiring tissues. Flat portion of curve => O_2 loads onto Hb in lungs.

- In the lungs where the partial pressure of oxygen is very high at about **12 kPa**, haemoglobin will be about **96%** saturated with oxygen – almost every haemoglobin molecule will be combined with its full complement of eight oxygen atoms.
- In an actively respiring muscle where the partial pressure of oxygen is very low at about **2 kPa**, the hemoglobin will be about **20%** saturated with oxygen – that is

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the haemoglobin is carrying only a quarter of the oxygen which it is capable of carrying.

- This means that haemoglobin coming from the lungs carries a lot of oxygen; as it reaches a muscle, it releases around three-quarters of it (i.e. $96\% - 20\% = 76\%$).
- This released oxygen diffuses out of the red blood cell and into the muscle where it is used in **respiration**.

TRY THIS QUESTION

Use the oxygen dissociation curve above to describe how the blood of a marathon runner carries and releases oxygen.

Answer

As a red blood cell passes through capillaries in the lungs, where the partial pressure of oxygen is very high (**12kPa**) its haemoglobin will **load up** oxygen i.e. pick up **4 oxygen molecules** and become about **96%** saturated. It is now called **oxyhaemoglobin** (HbO_8).

The red blood cell then goes to the heart from where it is pumped out to the leg of a marathon runner. The muscle is working hard i.e. respiring and using up a lot of oxygen. The partial pressure of oxygen in the muscle tissues is therefore very low (e.g. **2kPa**). As the red blood cell enters the capillaries within the muscle it finds itself in a region of very low partial pressures of oxygen and the haemoglobin therefore loses **76%** of its oxygen. In other words, it releases **76%** to the muscles and ends up only **20%** saturated. The muscle cells will use this oxygen for **aerobic respiration** which thus provides **ATP** so that it can carry on contracting and so that the athlete can carry on running.

The S-shaped curve

Why is the oxygen dissociation curve (ODC) S-shaped (sigmoid shaped)?

Each haemoglobin molecule can combine with a total of 4 O_2 molecules (one O_2 molecule for each of the haem groups).

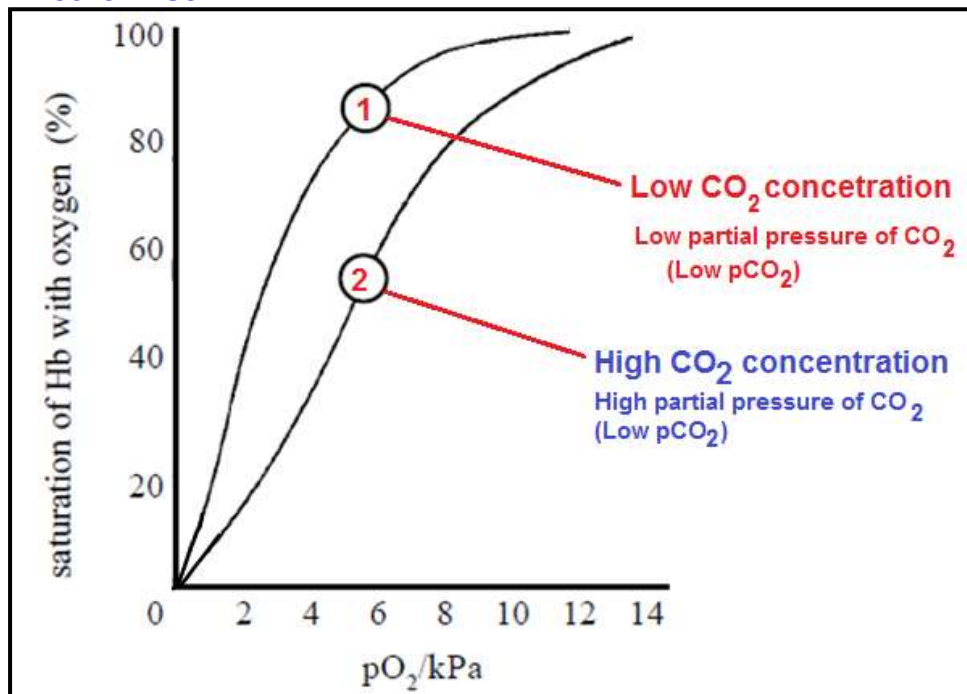
1. The first O_2 molecule combines relatively slowly with the first haem group. Thus, the first part of the ODC is not very steep.
2. However, when the first O_2 molecule combines with the first haem group, the whole haemoglobin molecule is slightly distorted.
3. The distortion makes it easier for the second and third O_2 molecules to bind to their haem groups. Thus, the ODC becomes much steeper.
4. The curve flattens off because it then becomes harder for the fourth (i.e. the last) O_2 molecule to combine with the fourth haem group.

- The shape of the hemoglobin dissociation curve reflects the way that oxygen atoms combine with hemoglobin molecules.
- Up to oxygen partial pressure of around 2 kPa, on average only one oxygen molecule is combined with each hemoglobin molecule.
- Once this oxygen molecule is combined, however, it becomes successively easier for the second and third oxygen molecules to combine, so the curve rises very steeply.
- Over this part of the curve, a small change in the partial pressure of oxygen causes a very large change in the amount of oxygen which is carried by the haemoglobin.

The Bohr shift

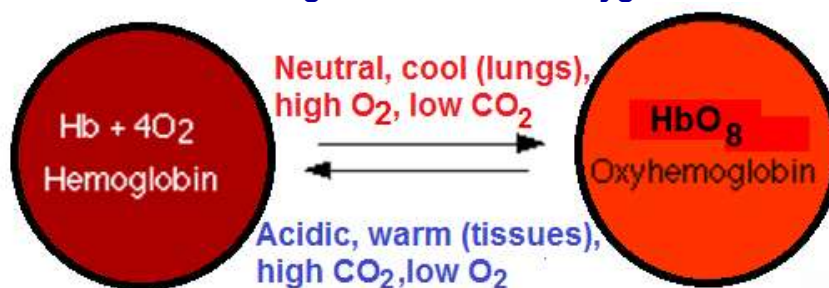
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- ❖ Bohr Effect is a property of hemoglobin as described by a scientist named Christian Bohr.
- ❖ The Bohr Effect is the right shift of the oxygen dissociation curve (ODC) in association with increased $p\text{CO}_2$ and H^+ concentration.
- The Bohr Effect states that the presence of a high partial pressure of carbon dioxide (high $p\text{CO}_2$) causes haemoglobin to release oxygen.
- The Bohr Effect also states that in regions with increased partial pressure of carbon dioxide the oxygen dissociation curve is shifted to the right.
- The physiological advantage of the shift is that in the presence of high concentrations of CO_2 and H^+ the O_2 affinity for Hb decreases causing the Hb to release O_2 .
- **Therefore the consequence of shifting ODC to the right is to facilitate O_2 unloading into respiring tissues.
- High $p\text{CO}_2$ are found in actively respiring tissues which need oxygen. This causes Hb to release oxygen even more readily than it would otherwise.



GRAPH: The Bohr Effect states that in regions with increased partial pressure of carbon dioxide the oxygen dissociation curve is shifted to the right

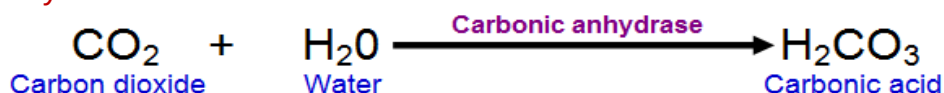
- Also according to the Bohr Effect an increase in blood carbon dioxide level, a decrease in pH (high H^+ concentration i.e. acidic conditions) or increased temperature causes haemoglobin to bind to oxygen with less affinity.



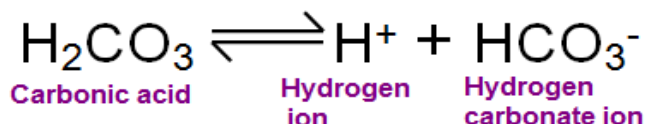
- The Bohr effect facilitates oxygen transport as haemoglobin binds to oxygen in the lungs, but then releases it in the tissues, particularly those tissues in most need of oxygen.

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- When a tissue's metabolic rate increases, its carbon dioxide production increases.
- The carbon dioxide is quickly converted into carbonic acid by the enzyme carbonic anhydrase:



- The carbonic acid dissociates into hydrogen ions (H^+) and hydrogencarbonate ions:



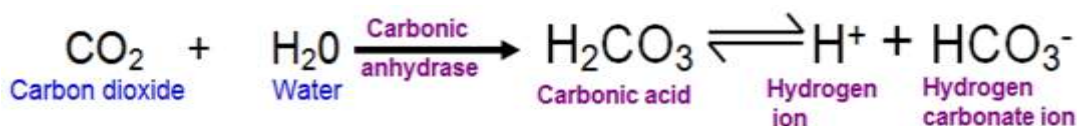
- The H^+ ions causes the pH of the tissue to decrease, and so increases the dissociation of oxygen from haemoglobin, allowing the tissue to obtain enough oxygen to meet its demands.
- Haemoglobin readily combines with the H^+ ions forming haemoglobinic acid, HHb. In so doing it releases the oxygen which it is carrying.
 - The hemoglobin 'mops up' the hydrogen ions which are formed when carbon dioxide dissolves and dissociates.
 - A high concentration of hydrogen ions means a low pH; if the hydrogen ions were left in solution, the blood would be very acidic. By removing the hydrogen (H^+) ions from solution, haemoglobin helps to maintain the pH of the blood close to neutral (pH 7.2). It is acting as a buffer.
 - The hydrogencarbonate ions diffuse out of the red blood cells into the plasma, thus leaving a shortage of negative ions in the red blood cells. To compensate for this, chloride ions (Cl^-) move from the plasma into the red cells to restore the electrical balance. This is known as the chloride shift.

Transport of carbon dioxide

The blood carries carbon dioxide in three ways:

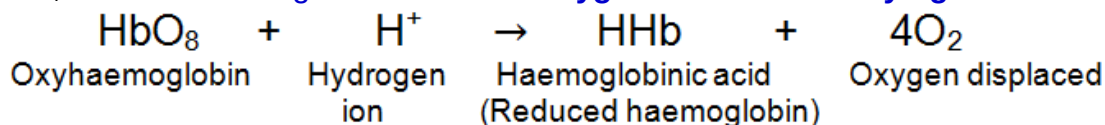
- 5% as carbon dioxide in solution in the plasma,
- 85% as hydrogencarbonate ions (HCO_3^-) in solution in the plasma,
- and 10% combined with red blood cells to form carbaminohaemoglobin. (The CO_2 combines with $-\text{NH}_2$ terminal of Hb of red blood cells to form carbaminohaemoglobin).

The remaining carbon dioxide enters red blood cells and is converted by the enzyme **carbonic anhydrase** to carbonic acid. This dissociates at once to form hydrogen ions and hydrogen carbonate ions.



As explained earlier, the H^+ ions causes the pH of the tissue to decrease, and so increases the dissociation of oxygen from haemoglobin, allowing the tissue to obtain enough oxygen to meet its demands.

- Haemoglobin (Hb) readily combines with the H^+ ions forming haemoglobinic acid, HHb. In so doing it releases the oxygen which it is carrying.



- The hemoglobin 'mops up' the hydrogen ions which are formed when carbon dioxide dissolves and dissociates. By removing the hydrogen (H^+) ions from

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solution, haemoglobin acts as buffer by maintaining the pH of the blood close to neutral (pH 7.2).

- The **hydrogencarbonate ions** **diffuse out of the red blood cells into the plasma**, thus leaving a **shortage of negative ions** in the **red blood cells**. To compensate for this, **chloride ions (Cl⁻)** **move from the plasma into the red blood cells to restore the electrical balance**. This is known as the **chloride shift**.
- When **blood reaches the lungs**, the relatively **low concentration of carbon dioxide in the alveoli** compared with that in the blood causes carbon dioxide to diffuse from the blood into the air in the alveoli **stimulating the carbon dioxide of carbaminohaemoglobin to leave the red blood cell** and **hydrogencarbonate** and **hydrogen ions** to **recombine to form carbon dioxide molecules once more**.
- This leaves the **hemoglobin molecules free to combine with oxygen**, ready to begin another circuit of the body.

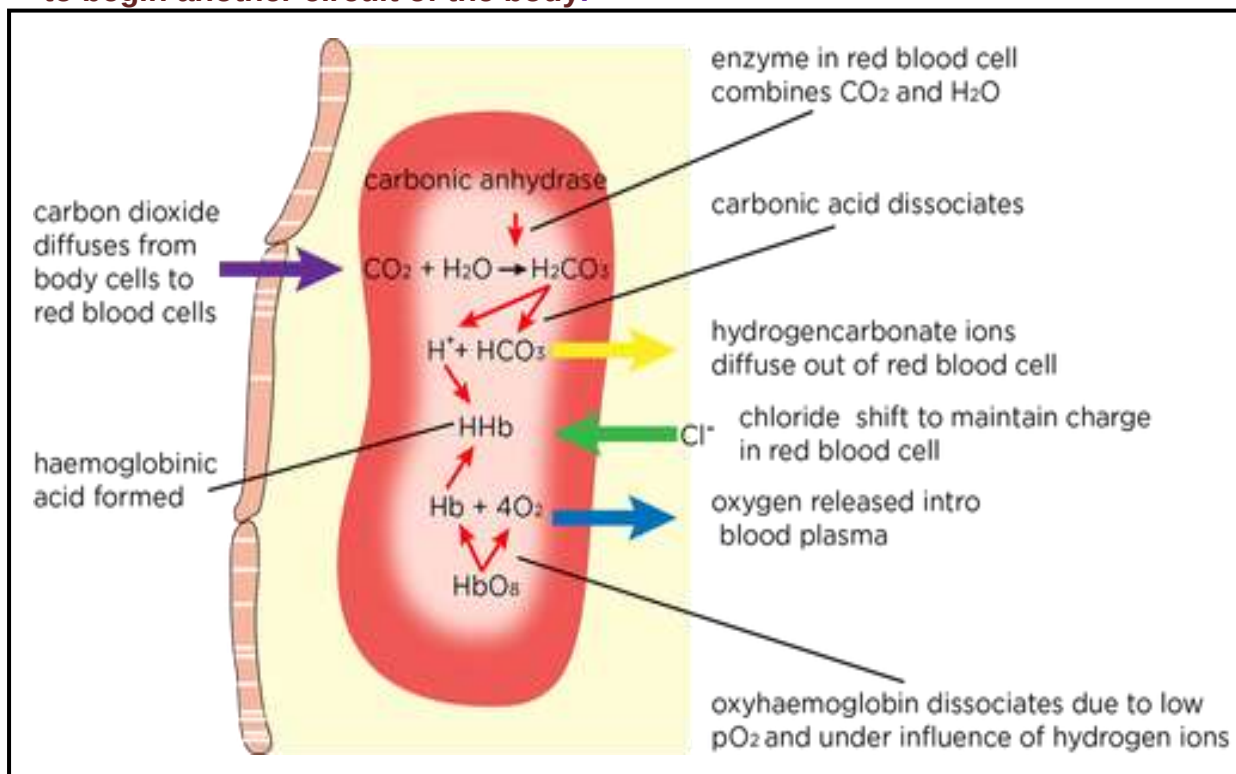


DIAGRAM: Transport of O₂ and CO₂

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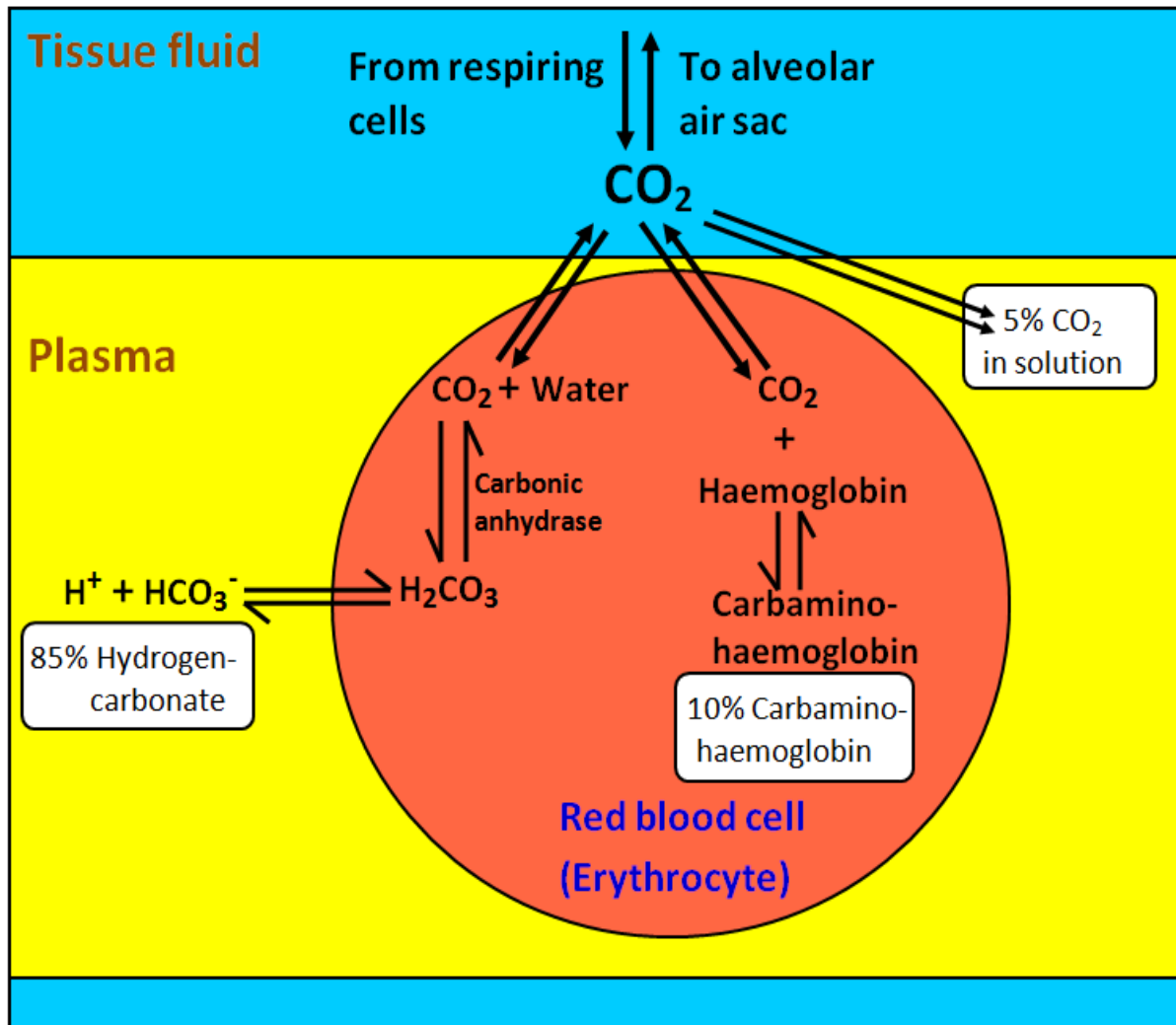
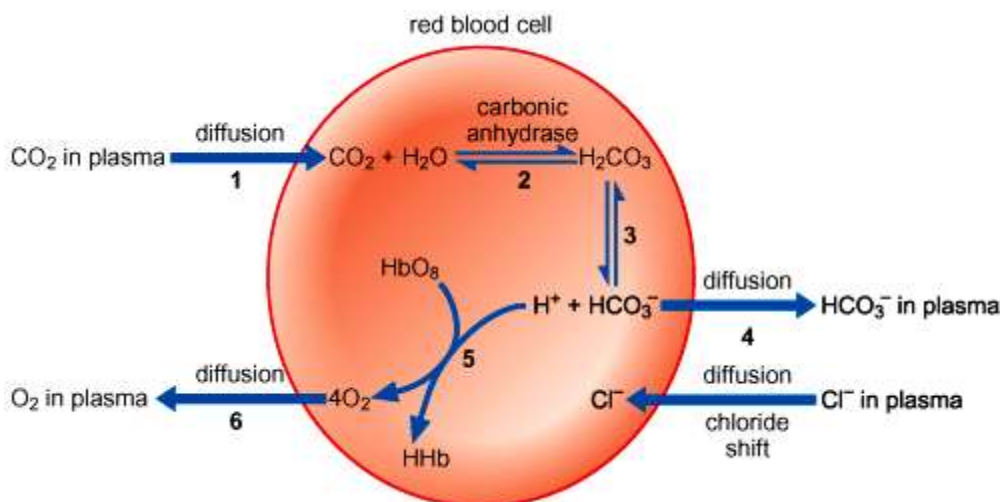


DIAGRAM: The blood carries carbon dioxide in three ways: partly as **undissociated carbon dioxide** in solution in the plasma, partly as **hydrogencarbonate ions (HCO₃⁻)** in solution in the plasma, and partly as **carbaminohaemoglobin** in red blood cells.

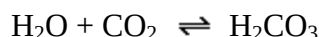
Try this question

Use the following diagram to describe the transport and effect of carbon dioxide in the blood.

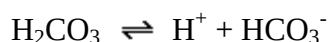


Answer

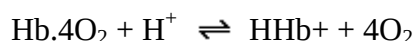
1. Carbon dioxide diffuses into the blood plasma and then into the red blood cells (erythrocytes).
2. Most CO_2 reacts with water in the red blood cells in the presence of the enzyme carbonic anhydrase to form carbonic acid in a reversible reaction:



3. Carbonic acid, H_2CO_3 , dissociates to form hydrogen ions and hydrogencarbonate ions. This is also a reversible reaction:



4. Negatively charged HCO_3^- ions diffuse out of red blood cells into the plasma. This is balanced by diffusion of chloride ions, Cl^- , in the opposite direction, maintaining the balance of negative and positive ions either side. This is called the '**chloride shift**'.
5. The dissociation of carbonic acid increases the acidity of the blood (decreases its pH). Hydrogen ions, H^+ , then react with oxyhaemoglobin to form haemoglobinic acid (HHb) and releasing bound oxygen in the process. This reduces the acidity of the blood. This buffering action allows large quantities of carbonic acid to be carried in the blood without major changes in blood pH.



($\text{Hb} \cdot 4\text{O}_2$ is sometimes written HbO_8 .)

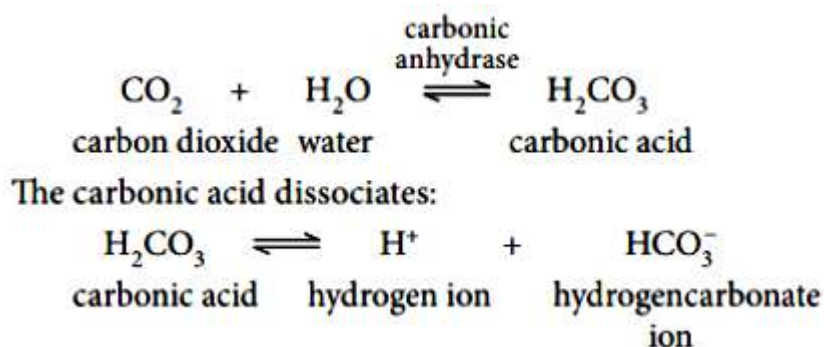
6. It is this reversible reaction that accounts for the Bohr Effect. Carbon dioxide is a waste product of respiration and its concentration is high in the respiring cell and so it is here that haemoglobin releases oxygen. The oxygen diffuses out of red blood cells into plasma and into respiring cells.

Summary

❖ **Haemoglobin (Hb) transport oxygen and carbon dioxide:**

1) In respiring tissues: The pCO_2 is high and the pO_2 is low.

- CO_2 from the cells diffuses into the plasma.
- CO_2 combines with $-\text{NH}_2$ terminal of Hb to form carbaminohaemoglobin. (10% carried this way)
- Most CO_2 combines with water (catalysed by carbonic anhydrase) to form carbonic acid which then dissociates into H^+ and HCO_3^- ions.



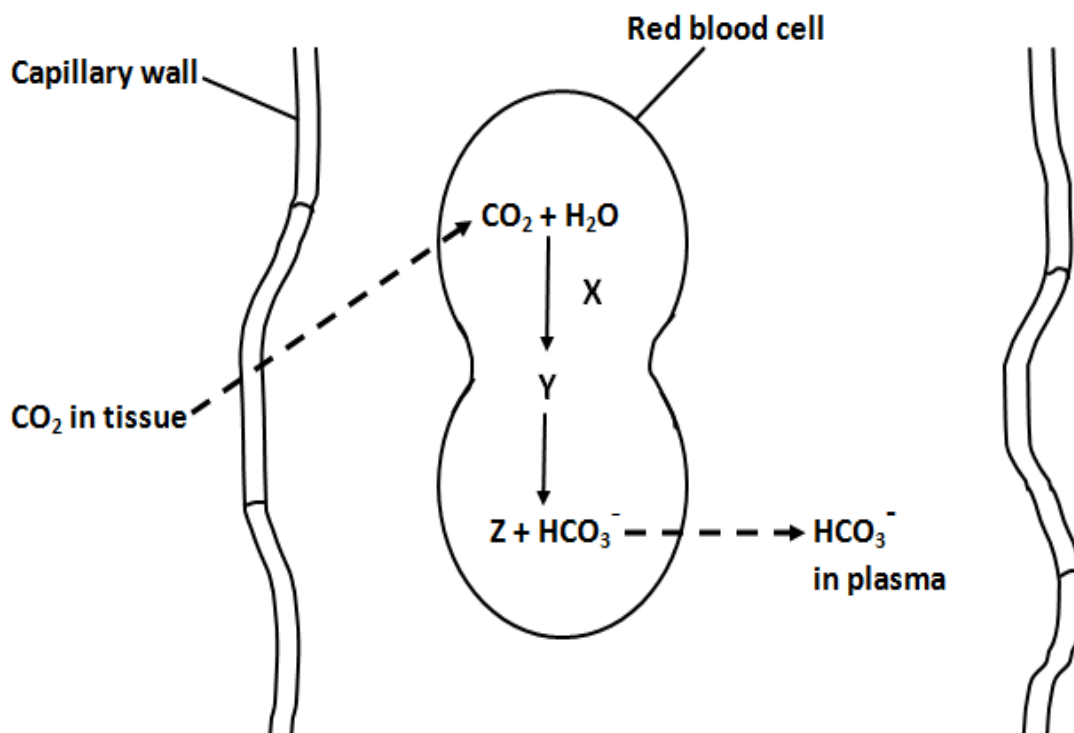
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- H^+ ions combine with Hb to form Haemoglobinic acid (HHb).
- 2) **In the lungs: The pO_2 is high and the pCO_2 low.**
- CO_2 in plasma diffuses from the blood into the alveoli and oxygen diffuses into the blood from the alveoli.
 - Carbaminohaemoglobin dissociates to form CO_2 and Hb
 - Hb then picks up O_2 , and HHb (haemoglobinic acid) dissociates to form H^+ and Hb.
 - The H^+ ions combine with HCO_3^- to form carbonic acid, which dissociates to form CO_2 and water (catalysed by carbonic anhydrase).
 - CO_2 diffuses into alveoli.
 - The presence of a high pCO_2 causes Hb to release oxygen. This is called the **Bohr Effect**. High pCO_2 are found in actively respiring tissues which need oxygen. This causes Hb to release oxygen even more readily than it would otherwise.

Try this question

Qn. Carbon dioxide is produced in tissues as a waste product of respiration. The majority of carbon dioxide is carried as hydrogencarbonate ions (HCO_3^-) in the plasma. The figure that follows shows the chemical pathway in which carbon dioxide is converted into HCO_3^- in a red blood cell. Identify the following:

- a) enzyme X
- b) substance Y
- c) ion Z [3]



THE BOHR SHIFT, ITS CAUSES AND CONSEQUENCES

- ❖ The **Bohr shift** of the **oxygen dissociation curve (ODC)** is either to the **right** or **left**.
- ❖ **Oxygen affinity** is the **tightness** with which Hb binds to O_2 .

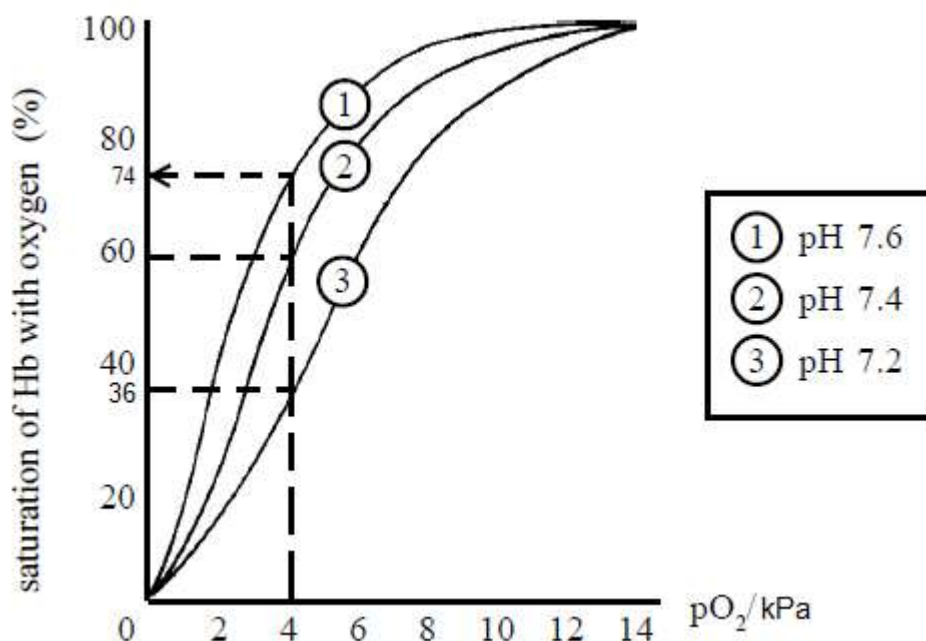
TRANSPORT SYSTEMS IN HUMANS

- Increased O₂ affinity means **increase in the tightness** with which Hb binds to O₂, therefore **LESS oxygen** is being **released** into the tissues=**LEFT SHIFT** of the ODC.
 - Decreased O₂ affinity means **decrease in the tightness** with which Hb binds to O₂, therefore **MORE oxygen** is being **released** into the tissues=**RIGHT SHIFT** of the ODC.
- ❖ The consequence of shifting ODC to **right** is to **facilitate O₂ unloading by Hb into respiring tissues**.
- Conditions/factors which cause decreased Hb affinity for O₂ cause right shift in ODC:
 - These factors that shift ODC to the right are: (mnemonics: **CADET**)
Increased:
 - 1) CO₂ (High pCO₂)
 - 2) Acid (High H⁺ concentration)
 - 3) 2,3-diphosphoglycerate
 - 4) Exercise
 - 5) Temperature
- Plus,
- HbS
 - Anaemia
 - Pregnancy
 - Post-acclimatisation to altitude
- Under these conditions oxygen is unloaded by Hb into respiring tissues
- ❖ The consequence of shifting ODC to the **left** is to **facilitate O₂ uptake by Hb from the lungs**.
- ❖ Conditions/factors which cause increased Hb affinity for O₂ cause left shift in ODC:
- These factors that shift ODC to the left are:
 - 1) Increase pH
 - 2) Reduced temperature
 - 3) Low pCO₂ (Reduced CO₂)
 - 4) Reduced 2,3-diphosphoglycerate
 - 5) Foetal Hb
 - 6) Myoglobin (Mb)
 - 7) Carboxyhaemoglobin
 - Under these conditions oxygen is taken up by Hb from the lungs (or from the placenta as is the case with foetal Hb).

The effect of various factors on the oxygen dissociation curve (ODC)

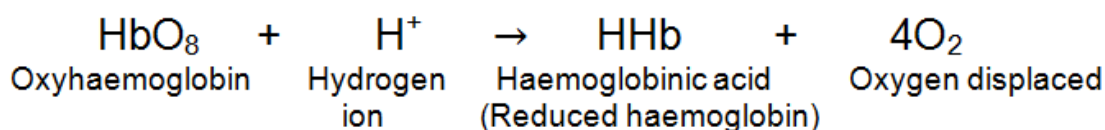
1. The effect of pH on the oxygen dissociation curve (ODC)

- ❖ The amount of oxygen carried by haemoglobin depends on **pH** as well as on **oxygen tension**.
- ❖ **In a more acidic environment oxygen dissociates from the haemoglobin more readily, whereas in a more alkaline environment association of oxygen with haemoglobin is favoured, forming more oxyhaemoglobin.** This results in the **Bohr effect** which is illustrated in the graph that follows.



GRAPH: The effect of pH on the ODC (The Bohr Effect)

- For example, imagine the blood is flowing through capillaries where the surrounding tissues have an oxygen tension of 4 kPa:
- At pH 7.6 the Hb would be 74% saturated
- But at pH 7.4 the Hb would be 60% saturated
- In other words, as pH decreases (becomes more acidic) the haemoglobin has become less saturated - it has given some of its oxygen to the tissues.
- At pH 7.2 (even more acidic conditions) the haemoglobin is only 36% saturated i.e. it has released even more oxygen to the tissues.
- ❖ **Increase in blood pH causes LEFT shift (increased affinity) in ODC:** as the acidity decreases (becomes more basic) the affinity will increase. This means that each Hb will hold on more tightly to the O₂ molecule. Therefore, there will be a **REDUCED** amount of O₂ going to the tissues leading to hypoxia (shortage of oxygen).
- ❖ **Decrease in blood pH causes RIGHT shift (decreased affinity) in ODC:** as the acidity increases the affinity of Hb for O₂ decreases and O₂ dissociates more readily from Hb; increase in acidity enhances the unloading of oxygen from hemoglobin (too much oxygen is going to the tissues).
- **So as a general rule, as pH decreases (as conditions become more acidic) haemoglobin becomes less saturated and gives up more oxygen.**
- Acidity can be defined as the concentration of hydrogen ions. **Hydrogen ions can reduce haemoglobin by attaching to the same sites as oxygen atoms and thus they compete with oxygen for these sites. In an acidic environment the higher concentration of hydrogen ions displaces oxygen from haemoglobin and thus improves oxygen release into the tissues.**

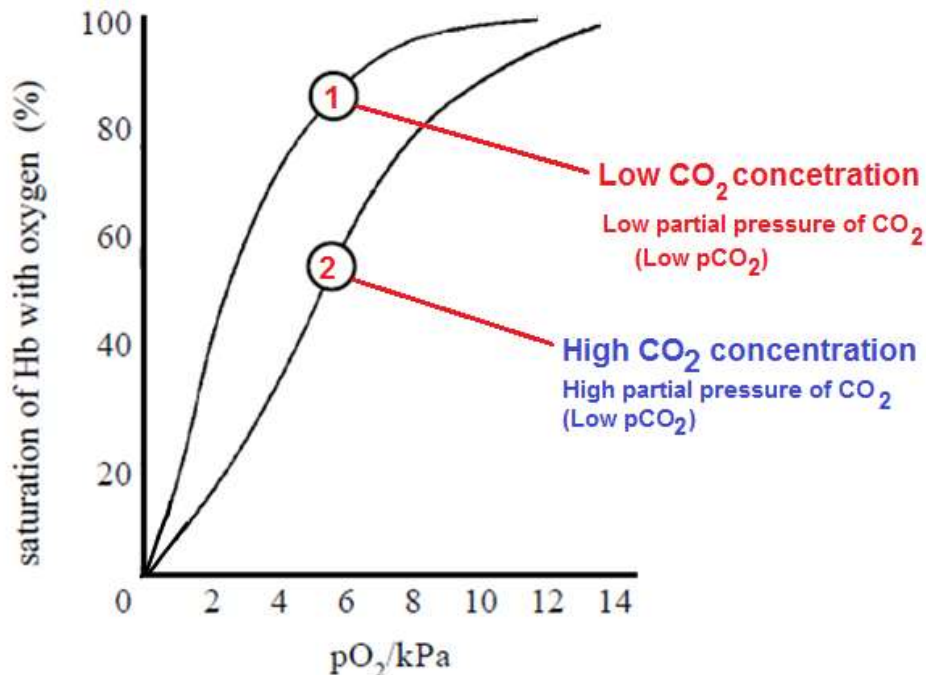


- This is perfect because respiring tissues are acidic as a result of the carbon dioxide and/or lactic acid produced in respiration. Carbon dioxide dissolves to form carbonic acid and this stimulates the release of oxygen from haemoglobin, thus allowing the tissues to keep respiring aerobically.

TRANSPORT SYSTEMS IN HUMANS

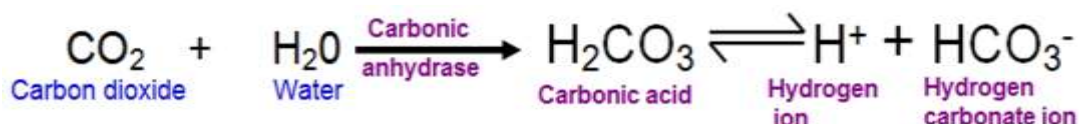
- The environment in the alveoli is less acidic (due to the lowering of carbon dioxide tension) and thus oxygen atoms predominate and tend to displace the hydrogen from the haemoglobin. This enhances the uptake of oxygen.
- In this way the haemoglobin is acting as a **buffer**, limiting the range of blood pH between 7.2 and 7.6. This is an example of homeostasis, in this case keeping the blood pH within optimum range.
- From the graph above it can be seen that: **As pH decreases, the ODC shifts to the right.** This is called the **Bohr Effect**.

2. The effect of Carbon dioxide on the oxygen dissociation curve (ODC)

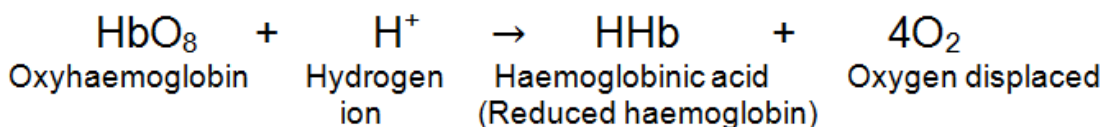


- The graph shows dissociation curves for haemoglobin at two different concentrations (partial pressures) of carbon dioxide.
- Curve 1 represents low CO₂ concentration.
- Curve 2 represents high CO₂ concentration.
- The shift of the curve to the right when haemoglobin is exposed to higher concentration of carbon dioxide is called the Bohr Effect.
- **The Bohr Effect states that in regions with increased partial pressure of carbon dioxide the oxygen dissociation curve is shifted to the right.**
- **The physiological advantage of the right shift is that in the presence of high concentrations of CO₂, the Hb affinity for O₂ decreases causing the Hb to release O₂.**
- **As CO₂ concentration increases, less oxygen combines with haemoglobin and the dissociation curve shifts to the right. In other words, as CO₂ concentration increases, more oxygen is released from haemoglobin.**
- **As CO₂ increases saturation of Hb decreases/haemoglobin releases more O₂. This provides respiring tissues with more O₂, allowing aerobic respiration to continue.**
- ❖ **Increase in CO₂ causes RIGHT shift in ODC i.e. decreased Hb affinity for O₂ and leads to a decrease in pH.**
 - The CO₂ enters red blood cells and is converted by the enzyme **carbonic anhydrase** to carbonic acid. This dissociates at once to form hydrogen ions and hydrogen carbonate ions.

TRANSPORT SYSTEMS IN HUMANS



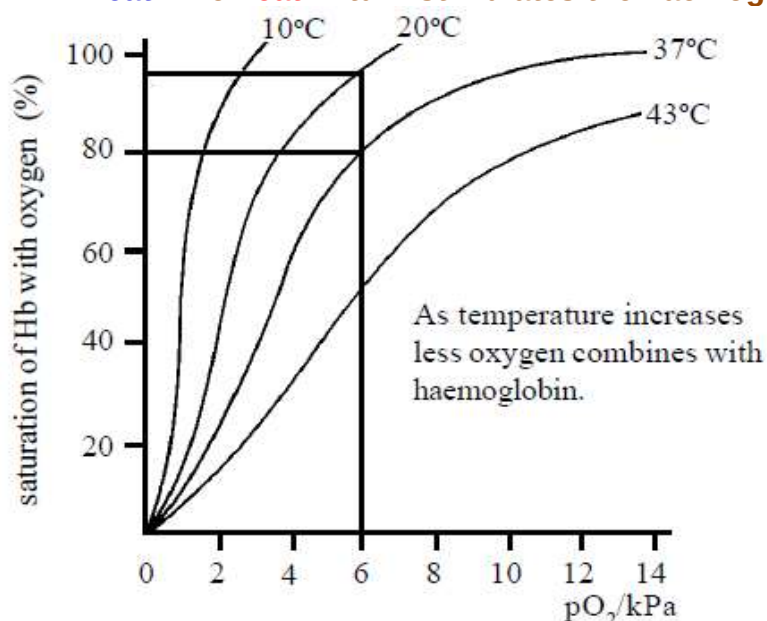
- The H^+ ions cause the pH of the tissue to decrease.
- Haemoglobin (Hb) readily combines with the H^+ ions forming haemoglobinic acid, HHb. In so doing it releases the oxygen which it is carrying allowing the tissue to obtain enough oxygen for aerobic respiration.



- The haemoglobin 'mops up' the hydrogen ions which are formed when carbon dioxide dissolves and dissociates. By removing the hydrogen (H^+) ions from solution, haemoglobin acts as buffer by maintaining the pH of the blood close to neutral (pH 7.2).
- The hydrogencarbonate ions diffuse out of the red blood cells into the plasma, thus leaving a shortage of negative ions in the red blood cells. To compensate for this, chloride ions (Cl^-) move from the plasma into the red blood cells to restore the electrical balance. This is known as the chloride shift.

3. The effect of temperature on the oxygen dissociation curve (ODC)

- The shift of the curve to the right when haemoglobin is exposed to higher temperatures is called the **Bohr effect**.
- ❖ As temperature increases, less oxygen combines with haemoglobin and the dissociation curve shifts to the right.
- ❖ In other words, as temperature rises, more oxygen is released from haemoglobin. Again, this is exactly what is needed. Heat energy is a product of metabolism. Active tissues such as contracting muscle fibres require large quantities of oxygen, and because of their activity release more heat. The heat in turn stimulates the haemoglobin to release more O_2 .



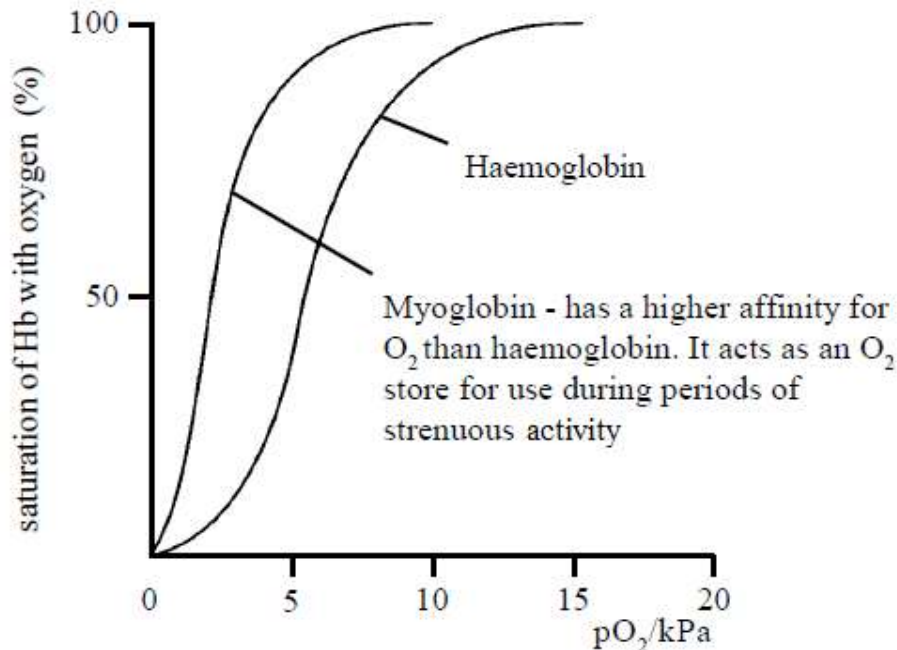
- For example, in tissues which have pO₂ = 6kPa:
 At 20°C the Hb would be 97% saturated
 At 37°C the Hb would be 80% saturated

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The difference - 17% - has been released to the respiring tissues.

*****Increase in temperature causes RIGHT shift of ODC i.e. decreased Hb affinity for O₂ occurs when temperature increases facilitating more oxygen release from Hb.** At the same pO₂, the oxygen saturation decreases.

4. Oxygen dissociation curves for haemoglobin and myoglobin



GRAPH: Myoglobin ODC shift to the **left** facilitates **oxygen loading** after haemoglobin has unloaded it.

- **Myoglobin** is a dark red respiratory pigment found in skeletal muscles.
- **Each myoglobin is made up of 1 polypeptide chain and can combine with one oxygen molecule to form oxymyoglobin.**
- **The oxymyoglobin is stable and will not release oxygen unless the partial pressure of oxygen around is very low. It therefore acts as a storage device for oxygen.**
- **Its oxygen dissociation curve lies to the left of that for haemoglobin so that it can pick up oxygen at partial pressures where haemoglobin is releasing it.**
- **The myoglobin will only release oxygen at very low partial pressures of oxygen i.e. when the oxygen concentration in the muscles drops very low - in other words, myoglobin acts as oxygen storage for use by muscles during periods of strenuous activity or exertion.**

❖ SUMMARY ON MYOGLOBIN:

- Found in skeletal & cardiac muscle.
- Has a higher affinity for O₂ as compared to Hb.
- Binds O₂ from arterial Hb.
- Releases it at O₂ tensions close to zero.
- Acts as O₂ storage (reservoir) for contracting muscles.
- Myoglobin ODC shift to the **left** facilitates **oxygen loading** after haemoglobin has unloaded it.

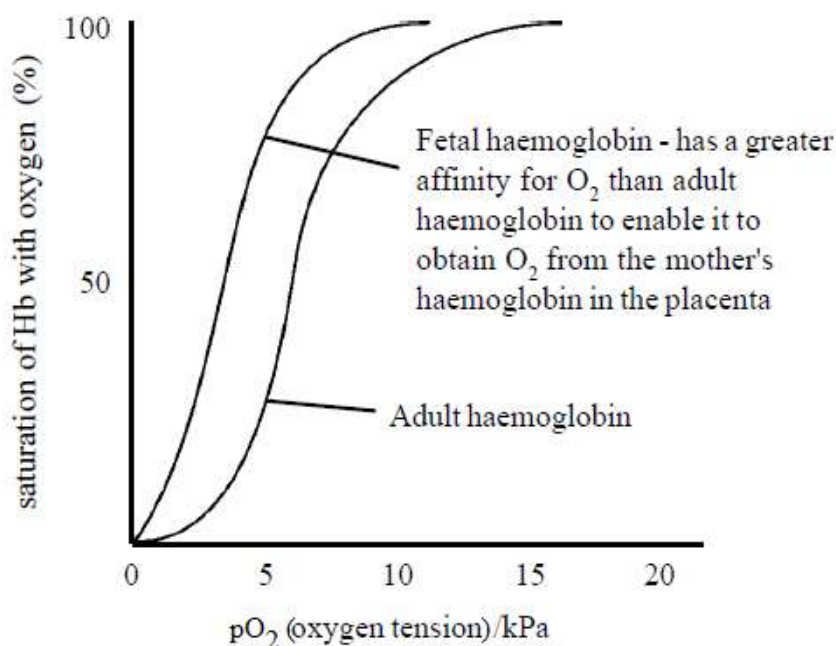
Question: State the differences between Myoglobin and haemoglobin.

TRANSPORT SYSTEMS IN HUMANS

BASIS FOR COMPARISON	HEMOGLOBIN	MYOGLOBIN
Number of chains	Haemoglobin has 4 chains of two different types- alpha and beta	It contains single polypeptide chains.
Type of structure	A tetramer.	A monomer.
Binds	Binds CO ₂ , CO, NO, O ₂ and H ⁺ .	Binds to O ₂ , tightly and firmly.
Their presence	Systemically all over the body.	In muscles cells.
Types of curve	Sigmoid binding curve.	Hyperbolic curve.
Also known as	Hb.	Mb.
Role	Haemoglobin is transported along with blood to whole body and carry oxygen.	Myoglobin supplies oxygen to muscles only, which is helpful at the starving time of oxygen.
Concentration in blood	High in RBC.	Low.

5. Oxygen Dissociation Curves of maternal and fetal haemoglobin

- Although a foetus has lungs, it does not use them until the moment of birth.
- Fetal haemoglobin takes oxygen from mother's haemoglobin across the placenta.
- Thus fetal haemoglobin must **pick up** oxygen at the oxygen tension at which maternal haemoglobin is **offloading** it.
- Thus fetal haemoglobin has a greater affinity for oxygen than adult haemoglobin.



GRAPH: ODCs of maternal and fetal haemoglobin. The ODC of maternal Hb is shifted to the right and the ODC of the foetal Hb is shifted to the left

Double Bohr Effect

What is the Double Bohr Effect?

– Situation in the placenta where the Bohr Effect operates in maternal and foetal circulations.

Explanation of the double Bohr Effect

In the MATERNAL PLACENTAL VILLI (INTERVILLOUS SINUSES)

- Increase in $p\text{CO}_2$ shifts ODC to **right** and facilitates O_2 unloading.

In the FOETAL CIRCULATION

- Decrease in $p\text{CO}_2$ shifts ODC to **left** and facilitates oxygen loading.

* Oxygen dissociation curves for maternal Hb & foetal Hb move apart

- maternal ODC = right shift

- foetal ODC = left shift

Structure of foetal and maternal haemoglobin

- ❖ Foetal red blood cells are nucleated and contain foetal haemoglobin, which is a conjugated protein with four haem prosthetic groups linked to two α and two γ polypeptide chains. Thus their different affinities for oxygen have a structural basis. One difference between them is shown below.
- ❖ Sequence of amino acids in part of adult haemoglobin
 - Phe – Ala – **Thr** – Leu – Ser – Glu – Leu – His – Cys –
- ❖ Sequence of amino acids in corresponding part of fetal haemoglobin
 - Phe – Ala – **Gln** – Leu – Ser – Glu – Leu – His – Cys –
- ❖ The difference in the sequence of amino acids results in different binding sites for oxygen and, in turn, different oxygen dissociation curves.
- ❖ Another difference between the blood of the mother and the foetus is that foetal red cells are nucleated. As birth approaches, foetal haemoglobin genes are switched off, adult haemoglobin genes are switched on and adult red cells containing adult haemoglobin are produced.

THE EFFECTS OF OTHER FACTORS ON BLOOD OXYGEN TRANSPORT

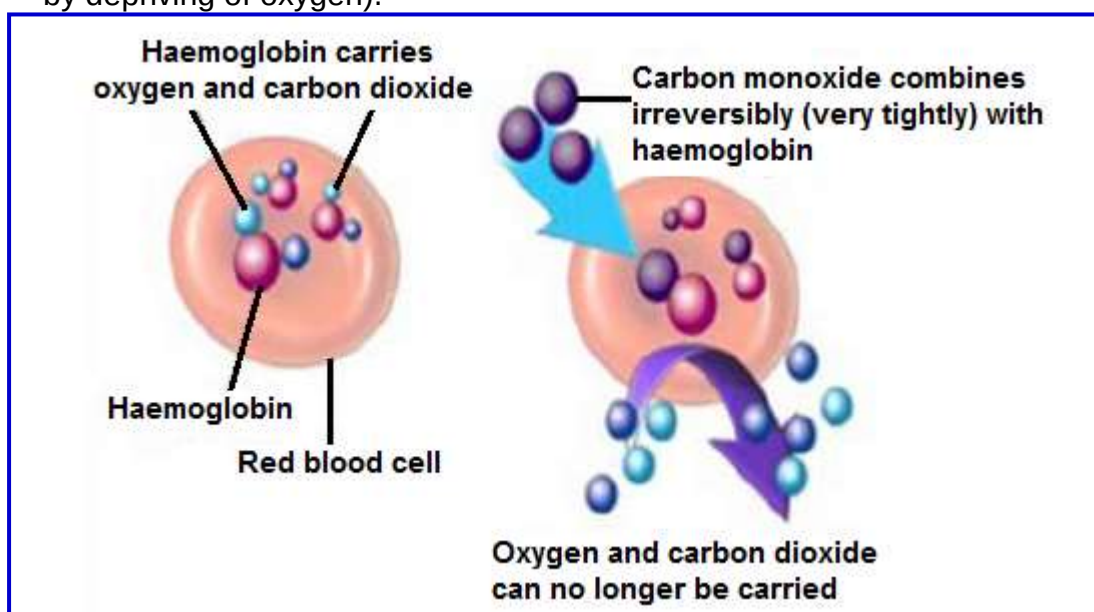
1. Presence of carbon monoxide (CO)



- ❖ Haemoglobin combines irreversibly with carbon monoxide to form carboxyhaemoglobin, a stable compound which reduces the efficiency of red blood cells to transport oxygen. This can cause death by asphyxiation (killing by depriving of oxygen).

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- Despite its almost perfect structure as an oxygen transporting molecule, hemoglobin has one property which can prove very dangerous. It combines very readily, and almost irreversibly, with carbon monoxide.
 - Carbon monoxide (CO) is formed when a carbon containing compound such as a fuel burns incompletely.
 - **Exhaust fumes** from cars contain significant amounts of carbon monoxide as does **cigarette smoke** as well as smoke from coal or charcoal **braziers** (*mbaura* in Shona).
 - When such fumes are inhaled the carbon monoxide readily diffuses across the walls of the alveoli, into blood, and into red blood cells.
 - Here it combines with the haem groups in the haemoglobin molecules, forming **carboxyhaemoglobin**.
 - Hemoglobin combines with carbon monoxide **250 times** more readily than it does with oxygen.
 - Thus, even if the concentration of carbon monoxide in air is much lower than the concentration of oxygen, a high proportion of hemoglobin will combine with carbon monoxide.
 - Moreover, carboxyhaemoglobin is a **very stable** compound; the carbon monoxide remains combined with the hemoglobin for a long time.
 - The result of this is that even relatively low concentrations of carbon monoxide, as low as 0.1% of the air, can cause death by **asphyxiation** (killing by depriving of oxygen).



- **Cigarette smoke** contains up to **5%** carbon monoxide. If you breathed in 'pure' cigarette smoke for any length of time you would die of asphyxiation.
- Even smokers who inhale also breathe in some normal air diluting the carbon monoxide levels in their lungs.
- Around 5% of the hemoglobin in a regular smoker's blood is permanently combined with carbon monoxide.
- This considerably reduces its **oxygen-carrying capacity**.
- ❖ **Treatment of carbon monoxide poisoning** involves administration of a mixture of pure oxygen and carbon dioxide: high concentrations of oxygen to favor the combination of hemoglobin with oxygen rather than carbon monoxide, and carbon dioxide to stimulate an increase in the breathing rate.

2.Disease.

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- This may cause increase in the body temperature (fever) and thus increase oxygen release to tissues.
- Anaemia will reduce the quantity of red blood cells and haemoglobin available to transport oxygen. Haemoglobin which is present in sickle cell anaemia has greatly reduced ability to transport oxygen.

3. Ecological factors.

- Animals which live in permanently reduced oxygen tensions tend to have haemoglobins which will pick up oxygen more readily at such tensions than normal haemoglobin.
- Thus, the dissociation curves of organisms such as mammalian parasites or mud dwelling worms usually lie to the **left** of the normal curve.

4. Body size.

- Small animals tend to have higher metabolic rates than large animals and thus their tissues require larger quantities of oxygen.
- Thus their dissociation curves will lie to the **left** of the normal curve, thus enhancing oxygen uptake and release.
- Their haemoglobins should however be equally effective at picking up oxygen.

5. High altitude



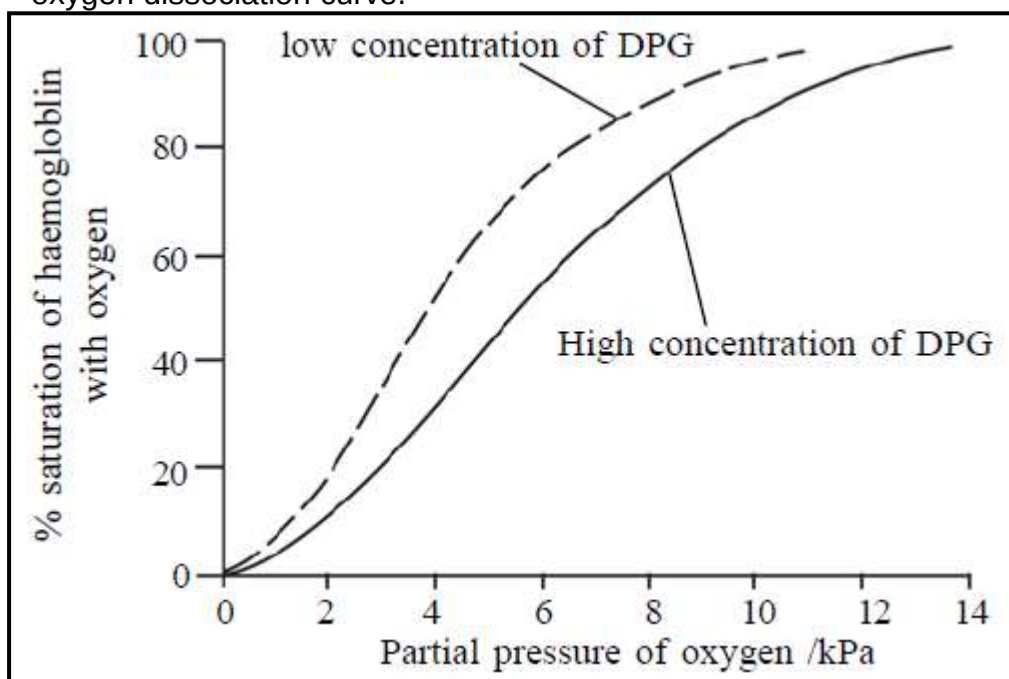
PICTURE: **Altitude sickness** is effects of **oxygen deficiency (hypoxia)** in the blood and tissues at **high altitudes**. The symptoms of altitude sickness include an increase in rate and depth of breathing, nosebleed, nausea, dizziness, weakness and mental disorientation.

- ❖ At high altitudes **the pO₂ is low**, which means **Hb will carry less oxygen** around the body. **This causes altitude sickness, which is the deficiency of oxygen (hypoxia) in the blood and tissues at high altitudes.**
- The **symptoms of altitude sickness** frequently begin with **an increase in the rate and depth of breathing**, and a general **feeling of dizziness, weakness, nosebleed, nausea, mental disorientation, hallucinations** and **euphoria** (a feeling of exaggerated elation).
- These symptoms can be easily **reversed** by **going down to a lower altitude**.
- Some people can quickly become very ill indeed. The arterioles in their brain dilate, increasing the amount of blood flowing into the capillaries, so that fluid

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begins to leak from the capillaries into the brain tissues. This can cause mental disorientation.

- Fluid may also leak into the lungs, preventing them from functioning properly.
 - Acute altitude sickness can be fatal, and a person suffering from it must be brought down to low altitude immediately, or given oxygen. However, if the body is given plenty of time to adapt, then most people can cope well at altitudes up to at least 5000 meters.
 - In 1979, two mountaineers climbed Mount Everest without oxygen, returning safely despite experiencing hallucinations and feelings of euphoria at the summit.
 - **In order to increase oxygen intake, populations that permanently live at high altitude areas (e.g. Andes or Himalayas) have adapted by developing:**
 - 1) **higher haemoglobin count,**
 - 2) broad chests providing **larger lung capacity,**
 - 3) **increased red blood cell count** and **greater number of mitochondria** to increase the efficiency of oxygen transport from lung to tissue.
 - 4) **thicker muscular wall of the right ventricle (therefore larger heart),** so more blood can be oxygenated.
 - 5) **increased concentration of 2,3-diphosphoglycerate (DPG) in their red blood cells which releases more oxygen at low pO_2 .**
- The graph below shows the effect of a high concentration of DPG on the oxygen dissociation curve.



- At low partial pressures of oxygen, the haemoglobin with high DPG in the red cells releases more oxygen than when there was a low DPG concentration.
- The haemoglobin releases more oxygen at low pO_2 just when it is most needed by respiring tissues of high altitude dwellers.
- It seems that high altitude dwellers are not genetically different from people who live at low altitudes, but rather that their exposure to low oxygen partial pressures from birth encourages the development of these adaptations from an early age.

THE HEART

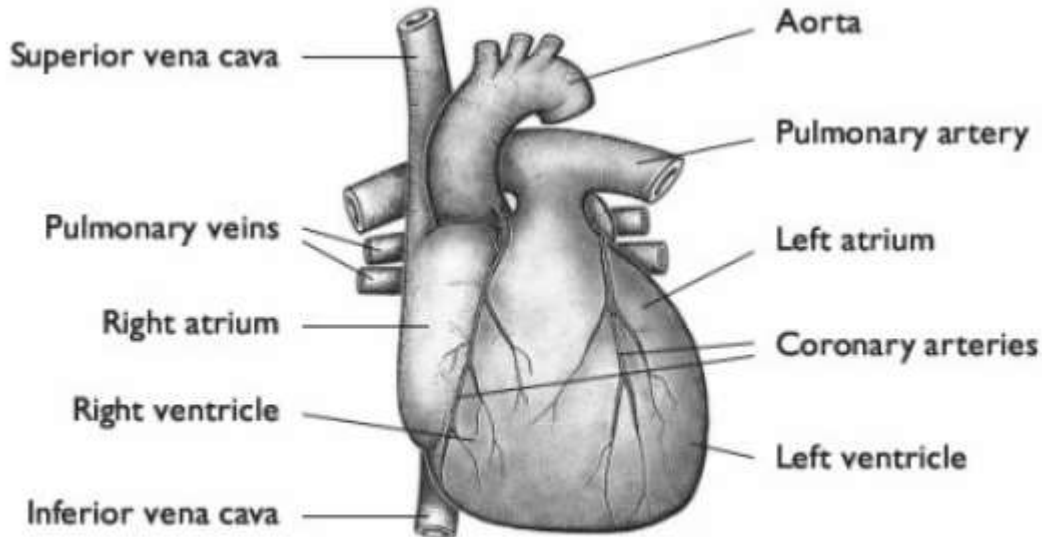
- ❖ The mammalian heart is a double pump:
 - The **right side** pumps **deoxygenated** blood at low pressure to the lungs.

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- The oxygenated blood then returns to the left side of the heart in the **pulmonary vein**.
- The **left side** pumps **oxygenated** blood at high pressure to the rest of the body in the aorta.
- The blood then returns to the right side of the heart in the **vena cava** to start the cycle again.

Structure of the mammalian heart

The external structure



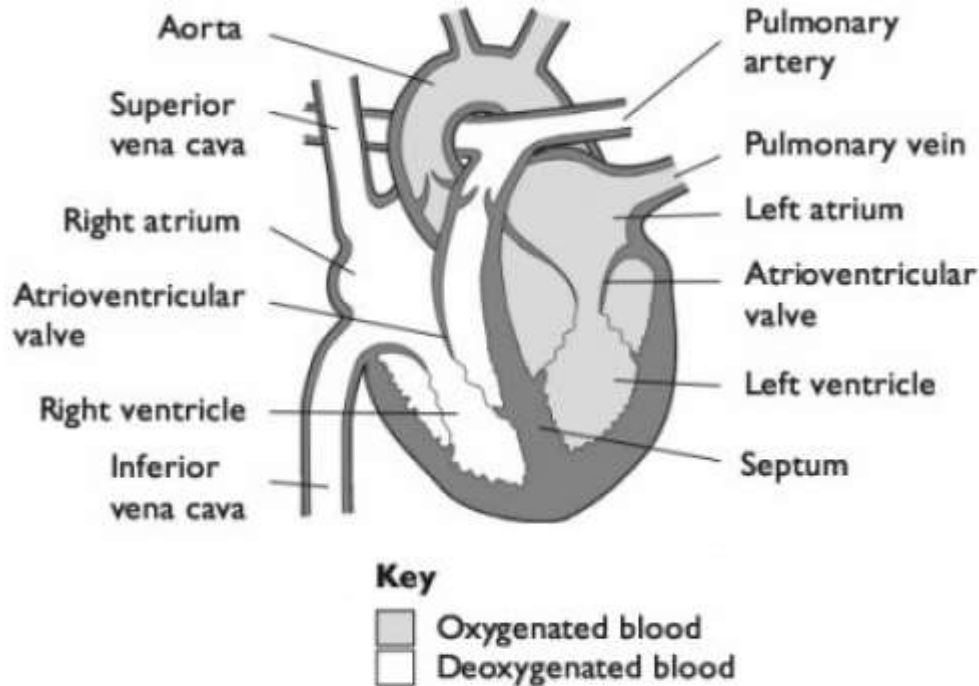
External view of a mammalian heart

- The mammalian heart is a hollow organ and is surrounded by a double membrane, called the **pericardium**.
- The space between the 2 membranes is filled with a watery fluid which prevents friction when the heart beats.
- A transverse and a longitudinal **groove** are visible on the surface of the heart. These grooves indicate the positions of the inner walls which divide the heart into **4 chambers**: 2 upper chambers (atria), and 2 larger lower chambers (ventricles).
- The **coronary arteries** and **veins** are clearly visible on the grooves.
- Entering the right atrium are the **superior vena cava** and the **inferior vena cava**.
- Entering the left atrium are the 4 **pulmonary veins**.
- From the upper central portion of the heart arise the **pulmonary artery** with a right and left branch and the **aorta** with its branches.

The Internal Structure

- A **muscular septum** divides the heart internally into a left and a right half. Each half is subdivided into 2 chambers, the atrium (reception) and the ventricle (pump chamber). Internally all 4 chambers have a smooth membranous lining, the **endocardium**.
- The atria and ventricle on the left side of the heart contain oxygenated blood, while those on the right side contain deoxygenated blood.
- The walls of the heart are made of cardiac muscle.

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Vertical section through a mammalian heart

- The walls of the ventricles are thicker than those of the atria as they have to produce more pressure.
- The wall of the left ventricle is thicker than the wall of the right ventricle as it has to produce enough pressure to move blood all around the body, not just to the lungs.

THE CARDIAC CYCLE

❖ The cardiac cycle is the sequence of one heartbeat.

- Involves repeated contraction and relaxation of heart muscle.
- Contraction is called **systole**.
- Relaxation is called **diastole**.
- During systole heart muscle contracts.
- During diastole heart muscle relaxes.

❖ Four stages of the cardiac cycle are

- Arterial diastole
- Arterial systole
- Ventricular systole
- Ventricular diastole.

1. Atrial diastole

- Atria and ventricles are both relaxed.
- Low pressure blood returning to heart through veins enters the atria.
- Oxygenated blood enters the left atrium.
- Deoxygenated blood enters the right atrium.
- At first the atrioventricular valves (bicuspid and tricuspid valves) are closed, but as atria fill with blood, pressure in them increases. Eventually it becomes greater than that in the relaxed ventricles and the valves are pushed open.

2. Atrial diastole

TRANSPORT SYSTEMS IN HUMANS

- When arterial diastole ends, the two atria contract simultaneously and push blood into the ventricles.

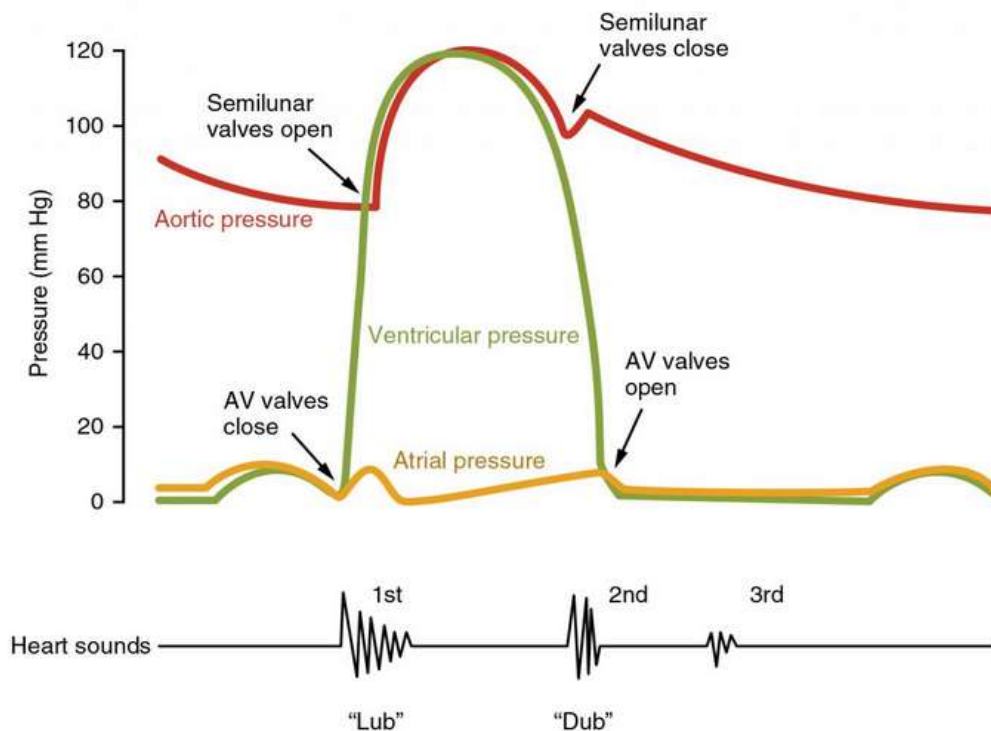
3. Ventricular systole

- Almost immediately (about 0.1 – 0.2 s later) the ventricles contract.
- Pressure in ventricles increases and closes atrioventricular valves preventing blood from returning into atria.
- The pressure forces open the semi-lunar valves of the aorta and pulmonary artery and blood enters these vessels. The first heart sound, “lub”, is produced by the closing atrioventricular valves during ventricular systole.

4. Ventricular diastole

- Ventricular systole ends and is followed by Ventricular diastole.
 - The ventricles relax.
 - The high pressure developed in the aorta and pulmonary artery tends to force some blood back towards the ventricles and this closes the semi-lunar valves of the aorta and pulmonary artery. Hence backflow of blood into the heart is prevented.
 - The second heart sound, “dub”, is produced by the closing semi-lunar valves during ventricular diastole.
- ❖ One complete heart beat consists of one systole and one diastole and lasts about 0.8 s.

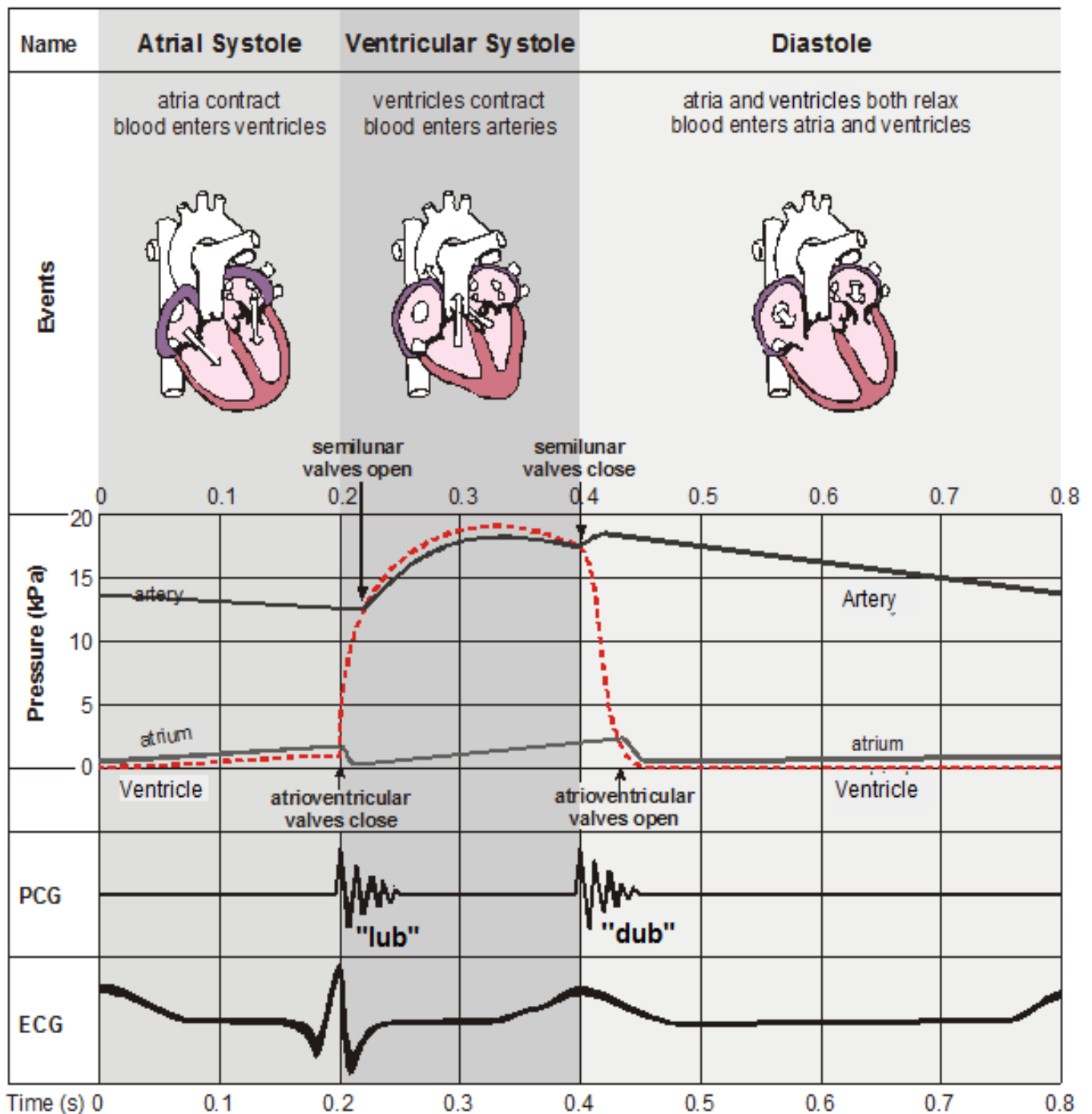
Graph of pressure changes during the cardiac cycle



- The events of the four stages of the cardiac cycle are shown in the diagram on the next page.
- The pressure changes show what is happening in each chamber.
- Blood flows because of pressure differences, and it always flows from a high pressure to a low pressure.
- So during atrial systole the atria contract, making the atrium pressure higher than the ventricle pressure, so blood flows from the atrium to the ventricle.

TRANSPORT SYSTEMS IN HUMANS

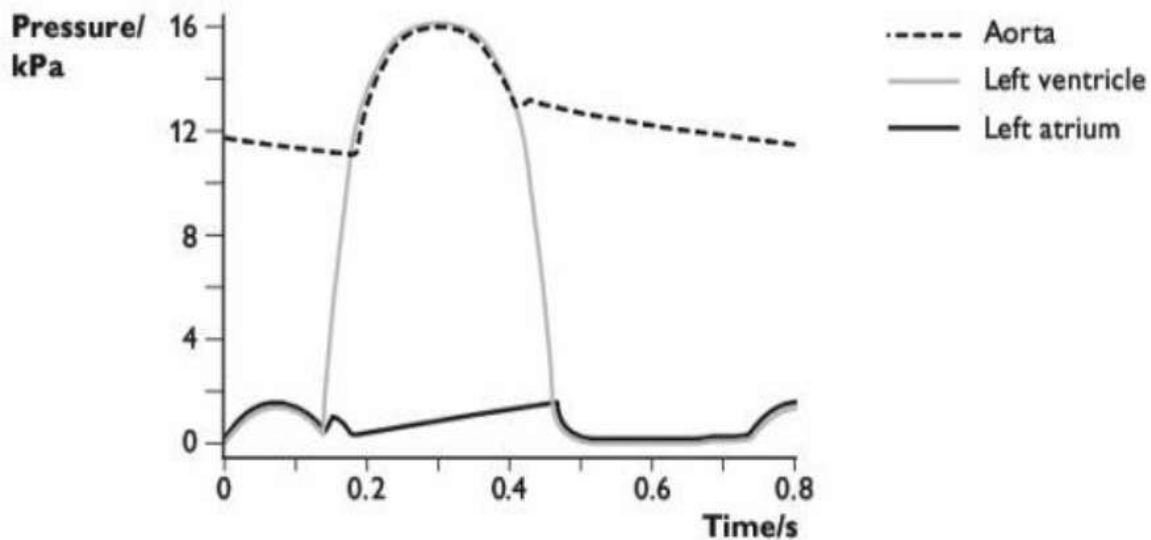
The artery pressure is higher still, but blood can't flow from the artery back into the heart due to the semi-lunar valves. The valves are largely passive: they open when blood flows through them the right way and close when blood tries to flow through them the wrong way.



- The **PCG (or phonocardiogram)** is a recording of the sounds the heart makes. The cardiac muscle itself is silent and the sounds are made by the valves closing. The first sound (lub) is the atrioventricular valves closing and the second (dub) is the semi-lunar valves closing.
- The **ECG (or electrocardiogram)** is a recording of the electrical activity of the heart. There are characteristic waves of electrical activity marking each phase of the cardiac cycle. Changes in these ECG waves can be used to help diagnose problems with the heart.

TRY THESE QUESTIONS

The diagram below shows pressure changes in the left atrium and left ventricle of the heart and the aorta during the cardiac cycle.



- (a) Calculate how many heart beats there will be in one minute. (2 marks)
- (b) (i) On the diagram, indicate the point at which the semilunar valves in the aorta snap shut. (1 marks)
- (ii) Explain what causes the semilunar valves to shut at this point in the cardiac cycle. (2 marks)
- (iii) On the diagram, indicate the period when the left ventricle is contracting. (1 mark)
- (iv) On the diagram, draw a line to show the changes in pressure in the right ventricle. (2 marks)
- (c) After the blood leaves the heart, it passes into the arteries. The blood pressure gradually reduces and becomes more steady as the blood passes through the arteries.

Explain what causes this reduction and steadying of the blood pressure.

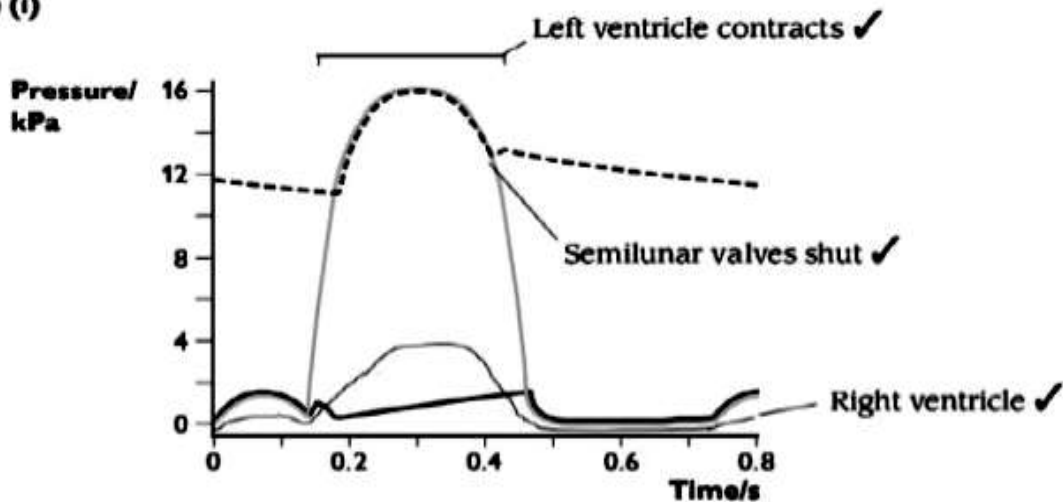
(2 marks)

(Total 10 marks)

(a) $60/0.75 \checkmark = 80$ beats per minute $\checkmark$

 Correct. 2/2

(b) (i)



 Correct. 1/1

(ii) They close when the pressure of the blood inside the arteries is higher than inside the ventricles $\checkmark$ — the blood therefore pushes down on the valves and makes them shut. $\checkmark$

 Correct. 2/2

(iii) See diagram.

 Correct. 1/1

(iv) See diagram.

 Correct 2/2

(c) As the blood is forced into the artery as the ventricle contracts, $\checkmark$ it pushes outwards on the artery wall, making the elastic tissue stretch. $\checkmark$ In between heart beats, the pressure of the blood inside the artery falls, and the elastic tissue recoils. $\checkmark$ So the wall keeps expanding and springing back. When it springs back it pushes on the blood in between $\checkmark$ heart beats, so this levels up the pressure changes.

 This answer explains very well why the blood pressure levels out. However, it does not mention the overall fall in blood pressure. All the same, this is a good answer which gets full marks. 2/2

Explain why the blood pressure in the left ventricles falls to zero in the cardiac cycle; but the lowest pressure recorded in the arteries is about 10kpa?

- Pressure falls to zero because all blood is expelled from ventricles;
- Pressure falls to 10kpa in arteries because of elastic recoil of the smooth muscles
- And the narrow diameter of the capillaries and arterioles
- This gives resistance to flow

WHAT CONTROLS HEART RATE?

1. **SAN (Pacemaker)** => Myogenic control
2. **Medulla** => Nervous control
3. **Hormones e.g. adrenaline, noradrenaline and thyroxine** => Hormonal control
4. **Other factors such as emotions, blood pH, blood temperature and mineral ions.**

Myogenic stimulation of heart rate

- Cardiac muscle is **myogenic**, that is, it contracts and relaxes automatically, without the need of stimulation by nerves.
- The rhythmic, coordinated contraction of the cardiac muscle in different parts of the heart is coordinated through electrical impulses passing through the cardiac muscle tissue.
- In the wall of the right atrium, there is a patch of muscle tissue called the **sinoatrial node (SAN)** or **pacemaker**. This has an intrinsic rate of contraction a little higher than that of the rest of the heart muscle.
- As the cells in the SAN contract, they generate action potentials (electrical impulses) which sweep along the muscle in the wall of the right and left atria. This causes the muscle to contract. This is **atrial systole**.
- When the action potentials reach the **atrioventricular node (AVN)** in the septum, they are delayed briefly. They then sweep down the septum between the ventricles, along the left and right **bundle of His** to the respective **Purkyne fibres**, and then up through the ventricle walls. This causes the ventricles to contract slightly after the atria. The left and right ventricles contract together, from the bottom up. This is **ventricular systole**.
- There is then a short delay before the next wave of action potentials is generated in the SAN. During this time, the heart muscles relax. This is **diastole**.

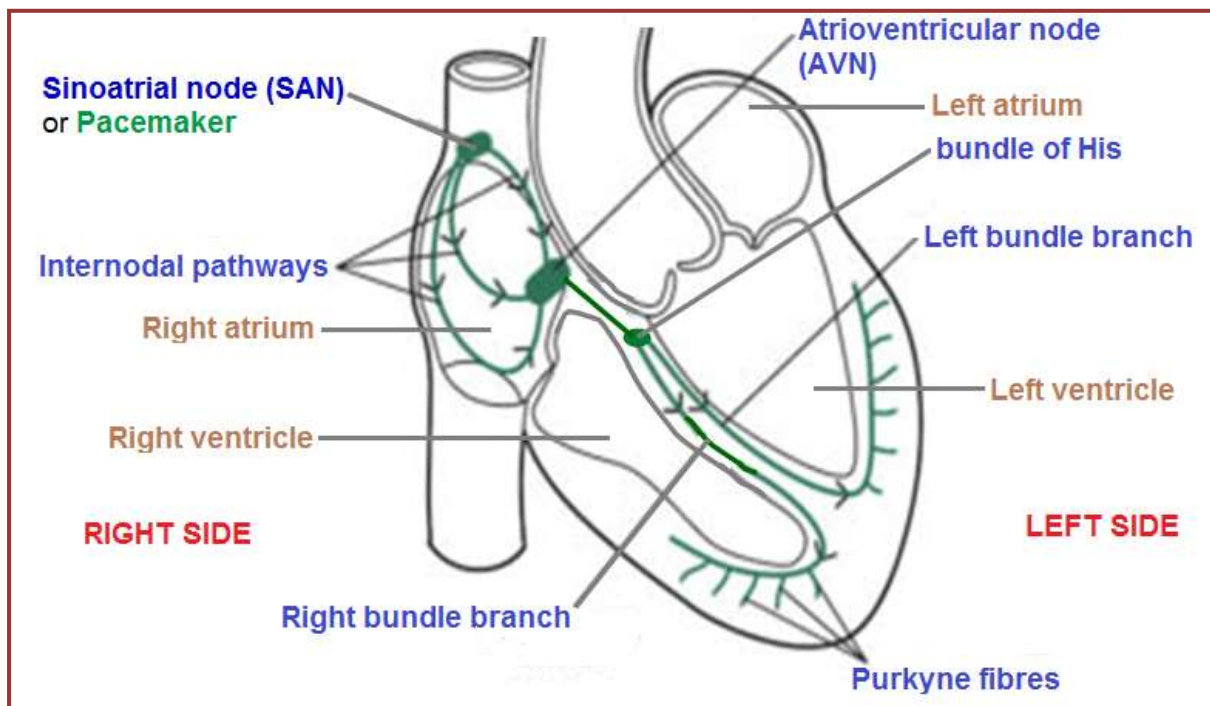


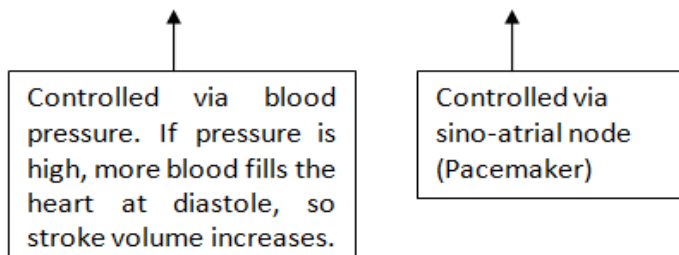
DIAGRAM: Cardiac conducting system.

Regulation of heart rate

TRANSPORT SYSTEMS IN HUMANS

- ❖ The **basic heart rate** is controlled by the **SAN** or **pacemaker** as explained earlier.
- ❖ The rate at which the heart beats and the volume of blood pumped at each beat (the stroke volume) can both be controlled. The product of these two is called the cardiac output – the amount of blood flowing in a given time:

Cardiac output = **stroke volume** x **heart rate**.



That is, **CO** = **SV** x **HR**

Try this question.

- (a) Use the formula: **Cardiac output** = **stroke volume** x **heart rate** to do calculations and complete the table that follows.

	Heart rate (beats/min)	Stroke volume (cm ³ /beat)	Cardiac output (cm ³ /min)
At rest	75	80	_____
During exercise	180	_____	21600

- (b) What is the effect of exercise on the cardiac output? What are the several benefits from this?

Answers

(a)

At rest: CO = SV x HR = 75 x 80 = 6000 cm³/min	During exercise: CO = SV x HR 21600 = SV x 180 SV = $\frac{21600}{180}$ SV = 120 cm³/beat
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	Heart rate (beats/min)	Stroke volume (cm ³ /beat)	Cardiac output (cm ³ /min)
At rest	75	80	6000
During exercise	180	120	21600

- (b) The cardiac output increases when the body exercises.

There are several benefits from this:

- to get oxygen to the muscles faster
- to get glucose to the muscles faster

TRANSPORT SYSTEMS IN HUMANS

- to get carbon dioxide away from the muscles faster
- to get lactic acid away from the muscles faster
- to get heat away from the muscles faster

Nervous control of heart rate

- But what makes the heart beat faster? What controls heart rate? Heart rate is clearly an involuntary process (you don't have to think about it), and like many involuntary processes (such as breathing, coughing and sneezing) it is controlled by a region of the brain called the **medulla** (a.k.a. **medulla oblongata**).
- That is, the function of the **medulla oblongata** is **the nervous control of the heart rate**.
- Nerves link the **medulla oblongata** with the heart as shown in the diagram below.

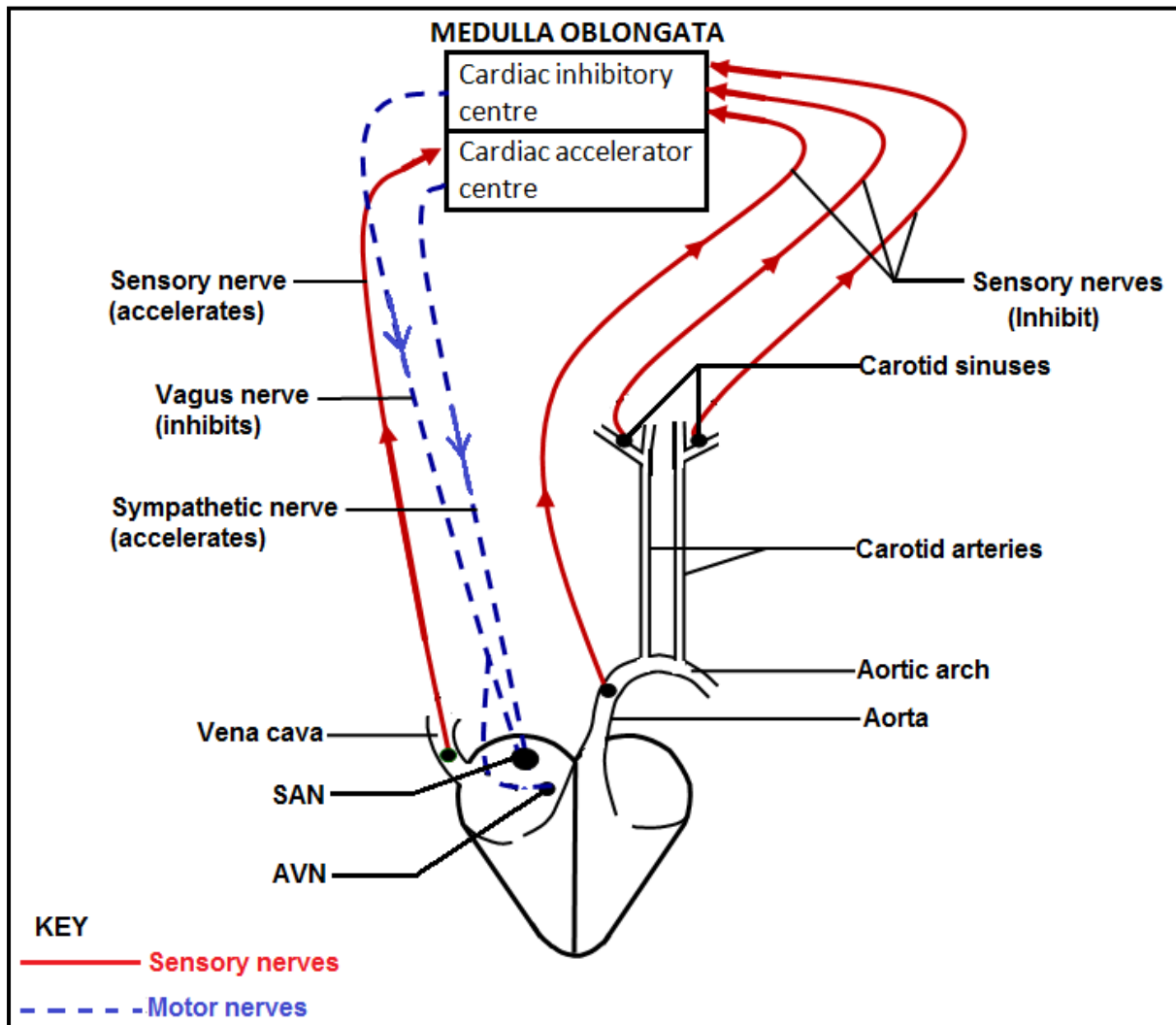


DIAGRAM: Nervous control of the heart rate. Neurones connect the heart to the cardiac inhibitory centre and cardiac accelerator centre in the medulla of the hindbrain.

- The nervous system is divided into a voluntary nervous system and an autonomic nervous system.
- The autonomic nervous system is itself divided into a sympathetic nervous system (SNS) and a parasympathetic nervous system (PNS).

TRANSPORT SYSTEMS IN HUMANS

- The SNS is generally associated with excitation of the body and preparing it for action, whereas the PNS has a relaxing influence. These both regulate the heart rate.
- The medulla oblongata has two regions affecting heart rate, the **cardiac inhibitory centre** which decreases the heart rate, and the **cardiac accelerator centre** which stimulates the heart rate.
- The coordinated activity of the cardiac inhibitory centre and the cardiac accelerator centre in the medulla oblongata controls the heart rate.
- Two parasympathetic nerves, called the **vagus nerves**, link the **cardiac inhibitory centre** to the **heart**.
- In the heart the vagus nerves branch and lead to the SAN, AVN, and the bundle of His.
- **Impulses that pass along the vagus nerve reduce heart rate.**
- Some nerves of the SNS start from the cardiac accelerator centre of the medulla oblongata and run parallel to the spinal cord and enter the SAN. Stimulation of these nerves results in an increase in the heart rate.
- Sensory nerve fibres from stretch receptors within the walls of the aortic arch, the carotid sinuses and the vena cava run to the cardiac inhibitory centre in the medulla oblongata
- Impulses received by the cardiac inhibitory centre from the aorta and carotids decrease heart rate.
- Impulses from the vena cava stimulate the cardiac accelerator centre to increase heart rate.
- As the volume of blood passing to these vessels increases so does the stretching of the walls of the vessels. This stimulates the stretch receptors and increases the number of nerve impulses transmitted to the centres in the medulla oblongata.
- For example, under strenuous exercise the body muscles contract strongly and this increases the rate at which blood in veins returns to the heart. As a result the walls of the vena cava are stretched by large quantities of blood and the heart rate is increased. At the same time the increased blood flow to the heart places the cardiac muscle of the heart under increased pressure. Cardiac muscle responds automatically (no nerves are involved) to this pressure by contracting more strongly during systole and pumping out increased volume of blood. In other words stroke volume is increased.
- The increased stroke volume stretches the aorta and carotids which in turn, via stretch reflexes, signal the cardiac inhibitory centre to slow down the heart rate. This is an automatic mechanism which prevents the heart working too fast, and enable it to adjust its rate in order to cope with the volume of blood passing through it at any given moment.

Hormonal control of heart rate

- ❖ Three hormones that have a direct effect on the heart rate are **Adrenaline**, **noradrenaline** and **Thyroxine**

Adrenaline and **noradrenaline**

- The **adrenal medulla** (of the **adrenal glands**) secretes the hormone **adrenaline** in large quantities and **noradrenaline** in small quantities. Noradrenaline is also secreted by the sympathetic nerves and acts as a neurotransmitter.
- Both these hormones increase heart rate, although adrenaline is more effective. The increased heart rate then causes the cardiac output and blood pressure to increase.

TRANSPORT SYSTEMS IN HUMANS

- These two hormones also prepare the body for action (the ‘flight or flight’ response).

Thyroxine

- Thyroxine, produced by the thyroid gland, raises basal metabolic rate. This in turn leads to greater metabolic activity, with greater demand for oxygen and production of more heat. As a result, vasodilation (dilation of blood vessels) followed by increased blood flow occurs, and leads to increased cardiac output.
- Thyroxine increase metabolism and therefore, there is need to pump blood faster to the respiring tissue to supply sufficient oxygen for the tissue metabolically.

Other factors that control heart rate

❖ **TABLE:** Factors other than nervous and hormonal factors which affect heart rate

Non-nervous stimulus	Effect on heart rate
High pH	Decelerates
Low pH (e.g. high CO ₂ levels, such as during active exercise)	Accelerates
Low temperature	Decelerates
High temperature	Accelerates
Mineral ions	Rate affected directly or indirectly

- ❖ **Emotions** also affect the cardiovascular centres in the medulla oblongata and impulses are transmitted to the heart causing changes in the heart rate. Examples of such emotions are embarrassment which causes blushing, anger which makes the skin to turn white, and also emotions generated by sounds and sights.

Effects of exercise on the circulatory (cardiovascular) system

❖ Short term effects of exercise on the circulatory (cardiovascular) system

1. During exercise the heart rate increases rapidly.
2. This provides the muscles with the necessary **oxygen** and **nutrients** to provide the muscles with **energy**.
3. During exercise, **cardiac output** is increased.

Cardiac output = **stroke volume** x **heart rate**.

- **Note:** **Stroke volume** is the amount of blood leaving each ventricle on each heart beat.

During exercise stroke volume increases because:

- more blood is sent back to the heart due to the muscles squeezing blood in the veins.
- as the heart fills up, it stretches.
- as the muscle fibres stretch, they contract more strongly, pumping out more blood.

❖ Long term effects of exercise on the circulatory (cardiovascular) system

1. **The heart muscle grows and gets stronger.**
 - Heart chambers get large and mass of heart muscle increases.
2. **Higher cardiac output.**

TRANSPORT SYSTEMS IN HUMANS

- The heart muscle will become more **efficient** in **heart rate** and **stroke volume**.
- 3. **Increase in blood vessels.**
- 4. **Increase in number and size of mitochondria in muscle fibres.**

Measuring Blood Pressure

- Blood pressure is the force developed by the blood pushing against the walls of the blood vessels.
- Blood pressure is measured in the **brachial artery in the arm** using a **sphygmomanometer**.
- Blood pressure is measured in millimetres of mercury, **mmHg**.
- There are two measurements of blood pressure:
 - **Systolic pressure** is the pressure produced by the **contraction of the ventricles**. (It is the **higher measurement** when the **heart beats**, pushing blood through the arteries), and
 - **Diastolic pressure** is the **pressure in the arteries when the ventricles relax**. (It is the **lower measurement** when the **heart rests** between beats).
- A young, fit person should have a blood pressure of about 120 over 80 (written, $\frac{120}{80}$), which means their systolic pressure is 120 mmHg and their diastolic pressure 80 mmHg.

High and Low Blood Pressure

- ❖ **Hypotension** (low blood pressure) is a condition in which blood pressure is abnormally low.
 - ❖ **Hypertension** (*high blood pressure*) is a condition in which blood pressure remains abnormally high (a reading of $\frac{140}{90}$ mm Hg or greater).
 - ❖ There are various factors that can increase blood pressure, including:
 - smoking
 - being overweight
 - atherosclerosis – narrowing and hardening of arteries
 - drinking a lot of alcohol
 - stress
 - ❖ Example: Smoking and the effect on blood pressure - Smoking increases blood pressure by raising the heart rate. Nicotine itself increases the heart rate and carbon monoxide reduces the oxygen-carrying capacity of the blood. It combines with haemoglobin in red blood cells, preventing oxygen combining with the haemoglobin. This causes an increase in heart rate to compensate for the reduced amount of oxygen carried in the blood.
- A balanced diet and regular exercise can reduce high blood pressure.
- ❖ Extremes of blood pressure can create problems. **High blood pressure (hypertension)** can cause:
 - kidney damage
 - burst blood vessels
 - damage to the brain, including stroke (cerebral haemorrhage) .
 - ❖ **Low blood pressure (hypotension)** can cause dizziness, fainting and poor blood circulation.